ggplot2

Customizando seus gráficos

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ggplot2

Visualizando conjuntos de dados

- Diamonds
- Gráfico de barras
- Histograma
- Geometrias para grande bases de dados
- Salvando gráficos

Diamonds

Entendendo mais um pouco deste banco

	carat	cut	color	clarity	depth	table	price	x	y	z
1	0.23	Ideal	E	SI2	61.5	55.0	326	3.95	3.98	2.43
2	0.21	Premium	E	SI1	59.8	61.0	326	3.89	3.84	2.31
3	0.23	Good	E	VS1	56.9	65.0	327	4.05	4.07	2.31
4	0.29	Premium	I	VS2	62.4	58.0	334	4.20	4.23	2.63
5	0.31	Good	J	SI2	63.3	58.0	335	4.34	4.35	2.75
6	0.24	Very Good	J	VVS2	62.8	57.0	336	3.94	3.96	2.48
7	0.24	Very Good	I	VVS1	62.3	57.0	336	3.95	3.98	2.47
8	0.26	Very Good	H	SI1	61.9	55.0	337	4.07	4.11	2.53
9	0.22	Fair	E	VS2	65.1	61.0	337	3.87	3.78	2.49
10	0.23	Very Good	H	VS1	59.4	61.0	338	4.00	4.05	2.39
11	0.30	Good	J	SI1	64.0	55.0	339	4.25	4.28	2.73
12	0.23	Ideal	J	VS1	62.8	56.0	340	3.93	3.90	2.46
13	0.22	Premium	F	SI1	60.4	61.0	342	3.88	3.84	2.33
14	0.31	Ideal	J	SI2	62.2	54.0	344	4.35	4.37	2.71
15	0.20	Premium	E	SI2	60.2	62.0	345	3.79	3.75	2.27

1.25 mm
0.010 ct

1.30 mm
0.011 ct

1.35 mm
0.012 ct

1.40 mm
0.013 ct

1.45 mm
0.014 ct

1.50 mm
0.015 ct

1.55 mm
0.016 ct

1.60 mm
0.019 ct

1.70 mm
0.020 ct

1.80 mm
0.025 ct

1.90 mm
0.030 ct

2.00 mm
0.035 ct

2.10 mm
0.040 ct

2.20 mm
0.045 ct

2.30 mm
0.050 ct

2.40 mm
0.060 ct

2.50 mm
0.065 ct

2.60 mm
0.070 ct

2.70 mm
0.075 ct

2.80 mm
0.085 ct

2.90 mm
0.10 ct

3.00 mm
0.11 ct

3.10 mm
0.12 ct

3.20 mm
0.13 ct

3.30 mm
0.14 ct

3.40 mm
015 ct

3.50 mm
0.16 ct

3.60 mm
0.18 ct

3.70 mm
0.20 ct

4.10 mm
0.25 ct

4.30 mm
0.30 ct

4.60 mm
0.35 ct

4.80 mm
0.40 ct

4.90 mm
0.45 ct

5.20 mm
0.50 ct

5.40 mm
0.60 ct

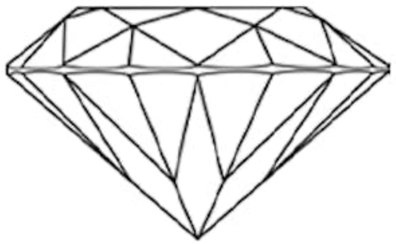
5.70 mm
0.70 ct

6.00 mm
0.80 ct

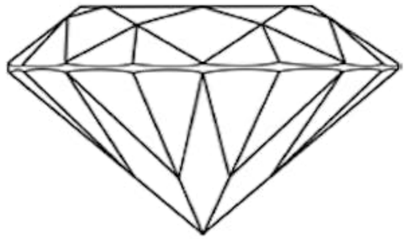
6.30 mm
0.90 ct

6.60 mm
1.00 ct

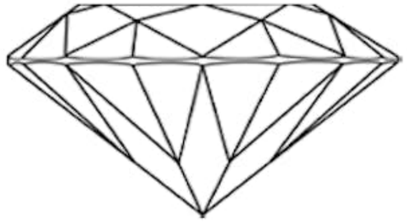
Ideal



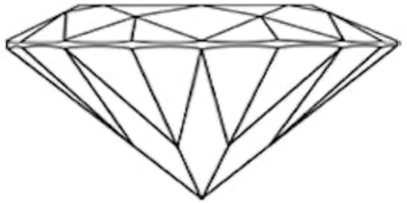
Premium



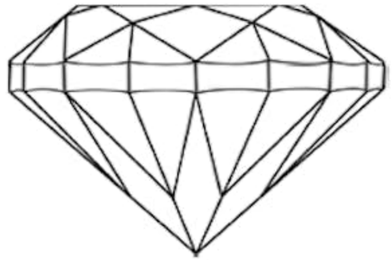
Very Good



Good



Fair



colourless
D-E-F

near colourless
G-H-I-J

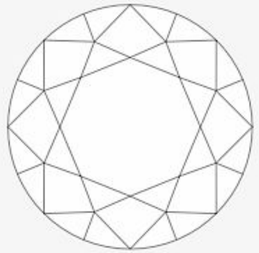
faint
K-L-M

very light
N-R

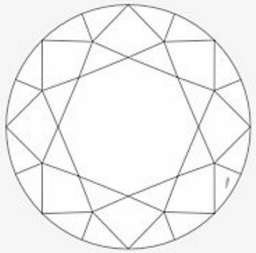
light
S-Z

Tiffany Standard

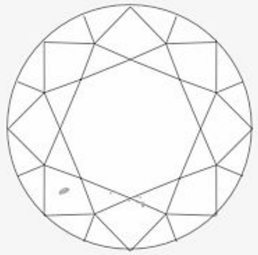
FL



IF



VVS1 VVS2



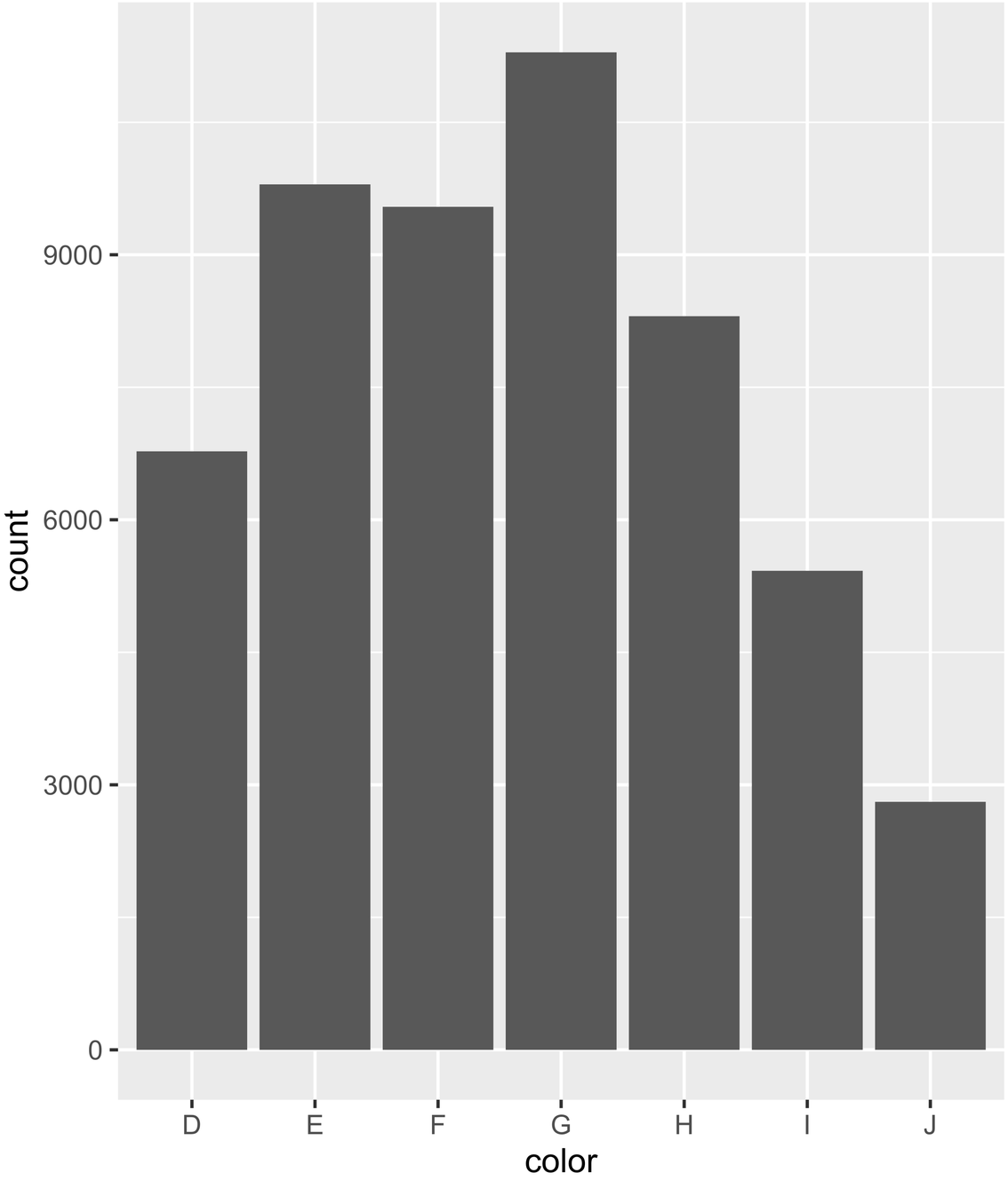
VS1 VS2

SI1 SI2

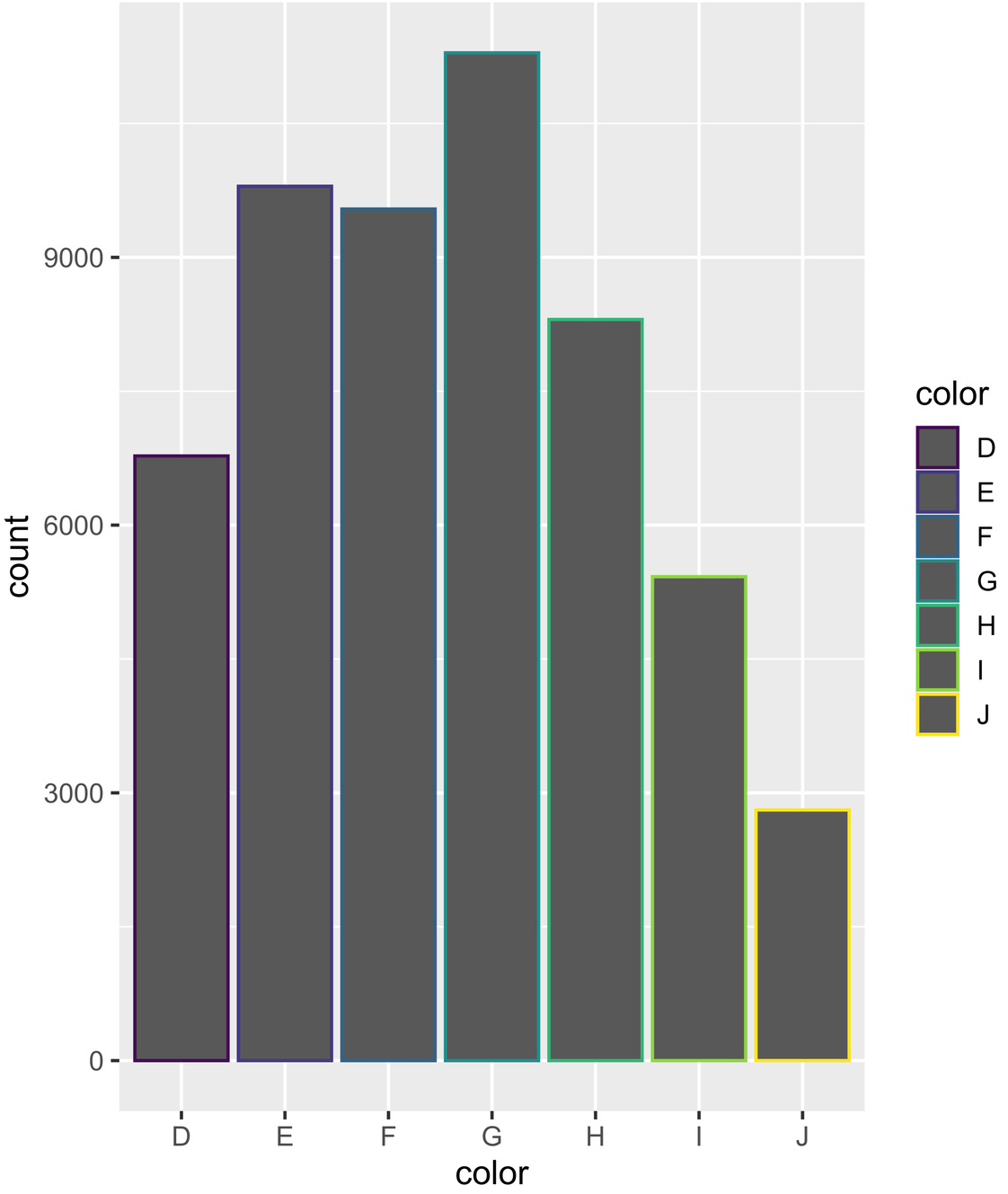
I1 I2 I3

Barras

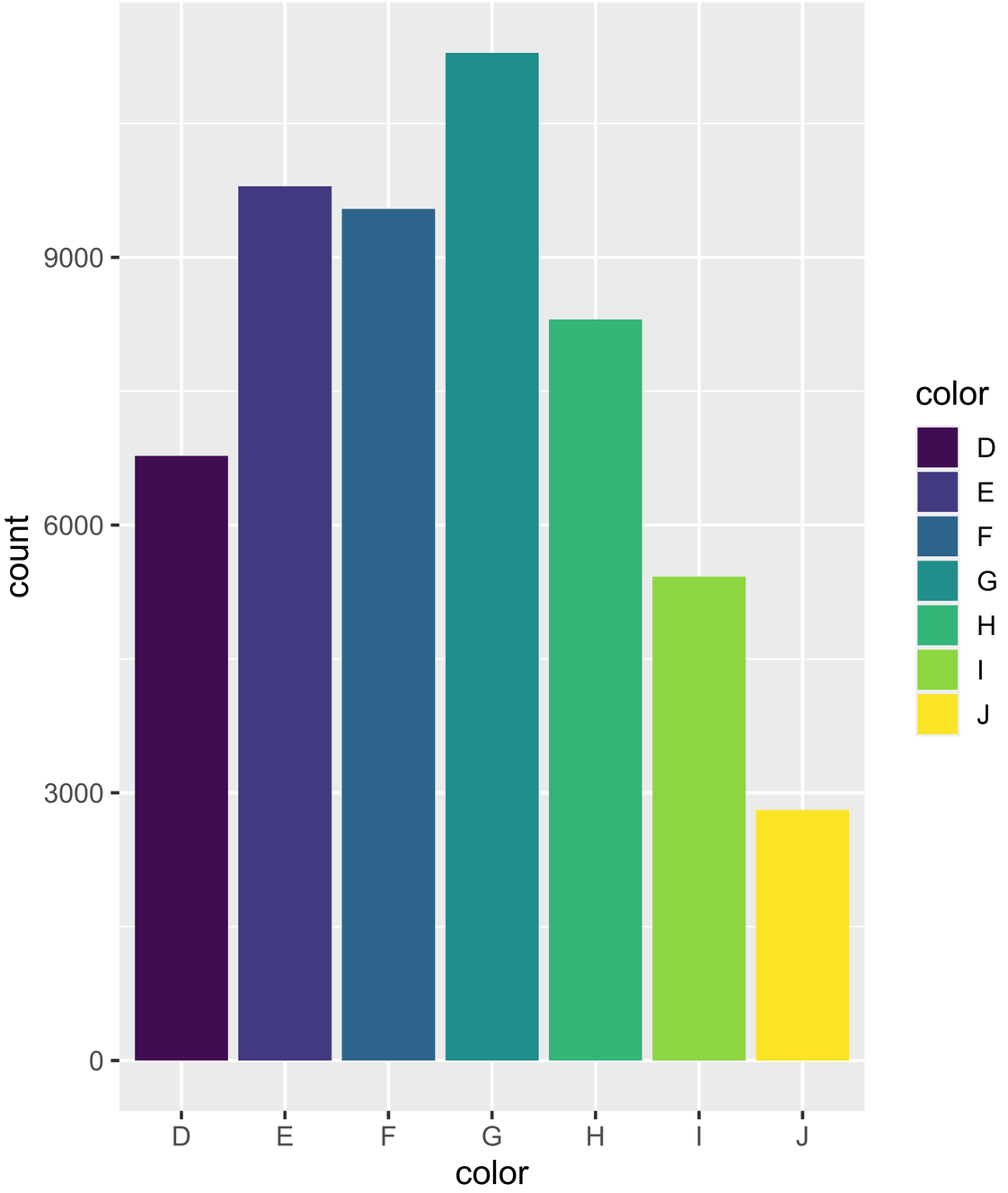
1 dimensão: categórico




```
ggplot( diamonds, aes(x = color) ) +
  geom_bar()
```

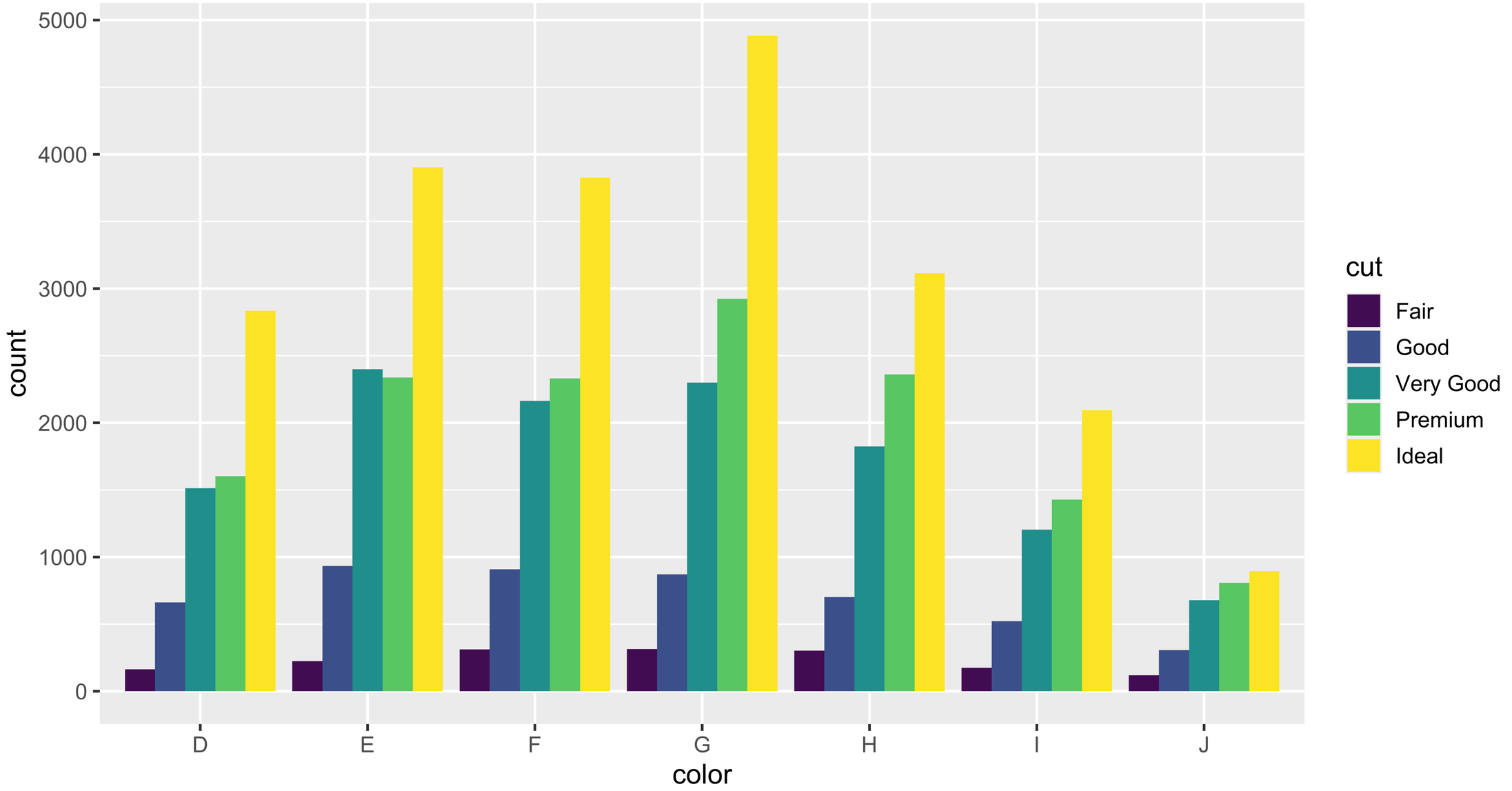



```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(color = color) )
```

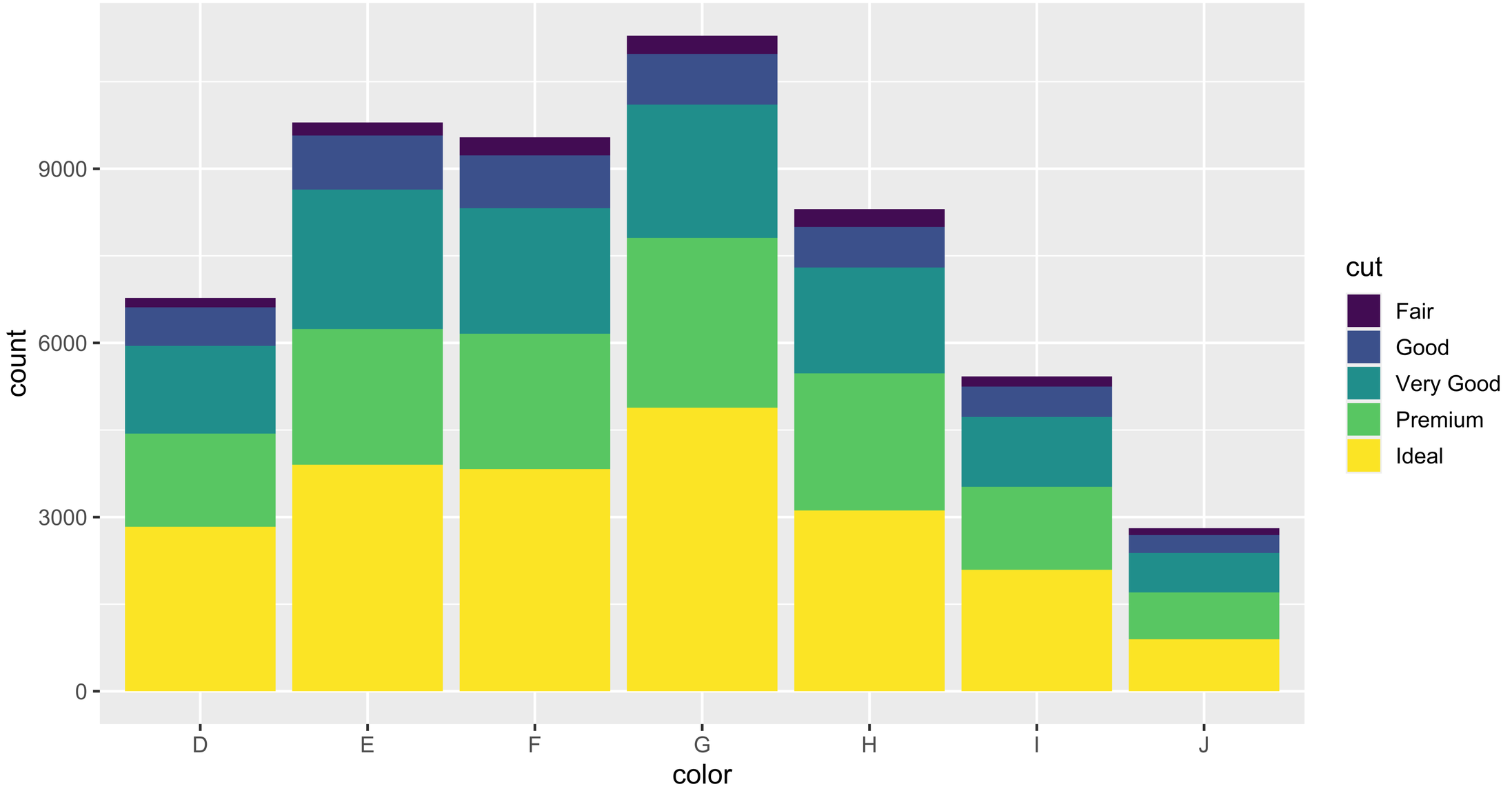



```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(fill = color) )
```

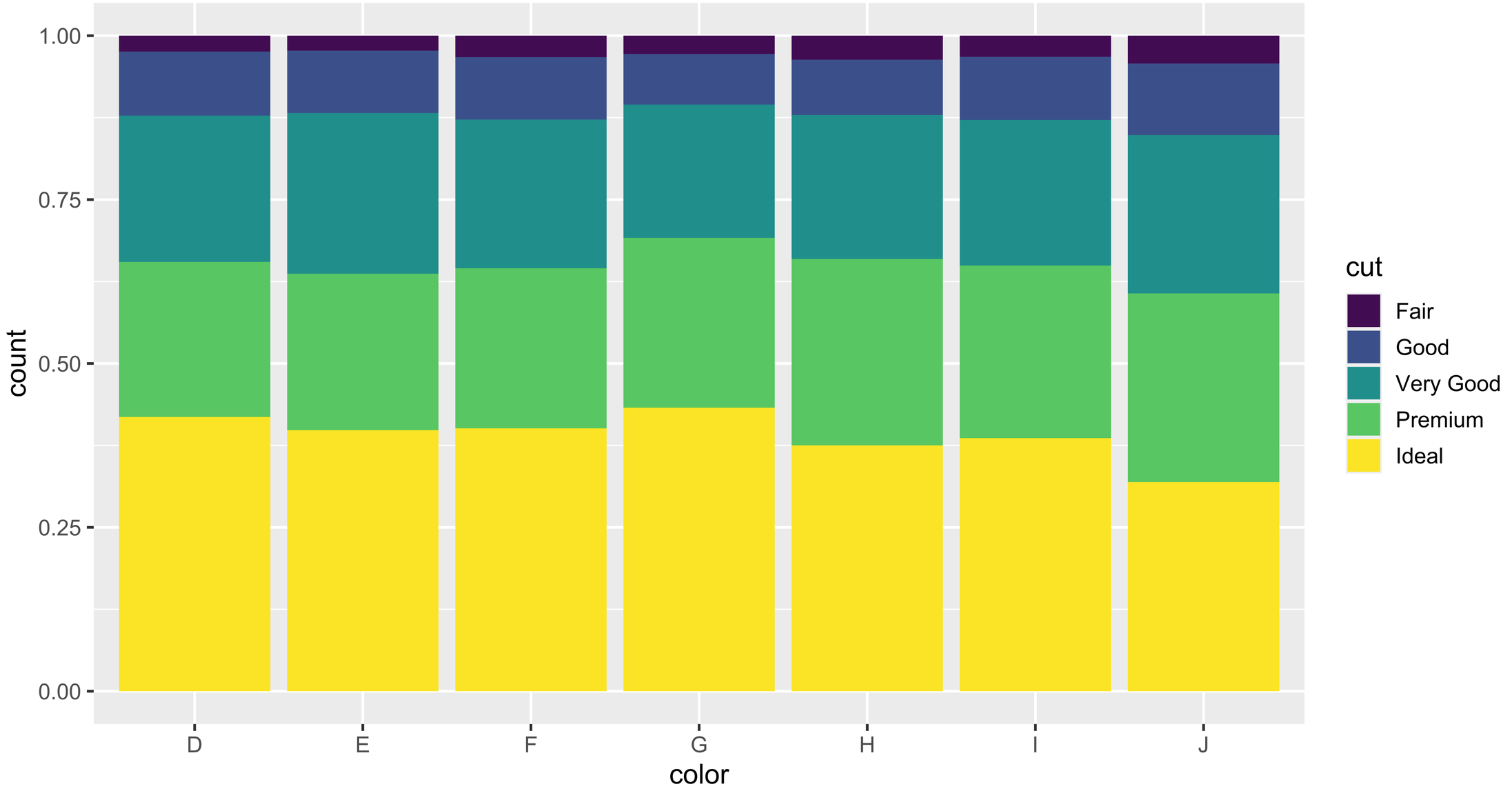
Ajustes de posicionamento



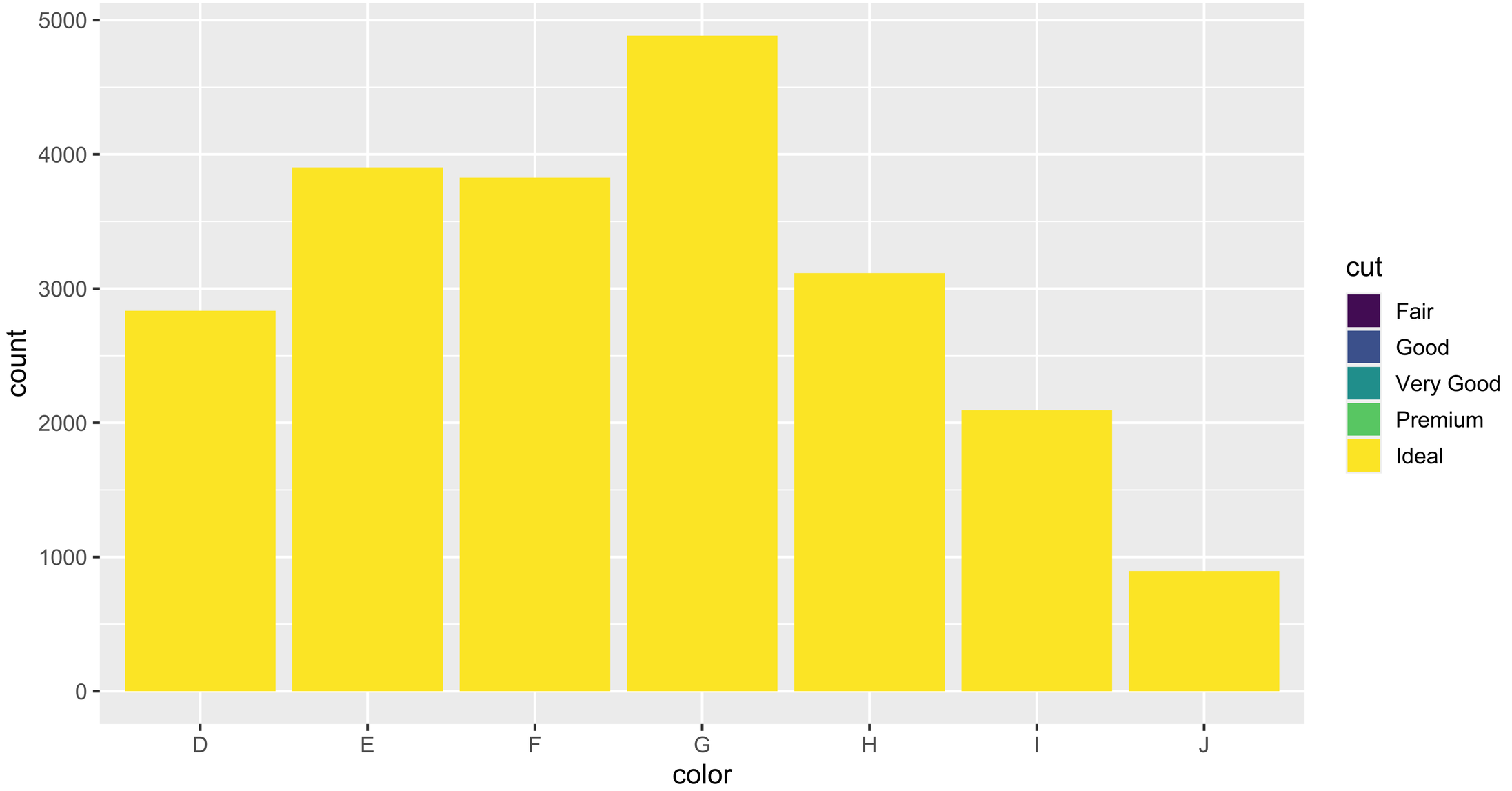

```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(fill = cut), position = "dodge" )
```



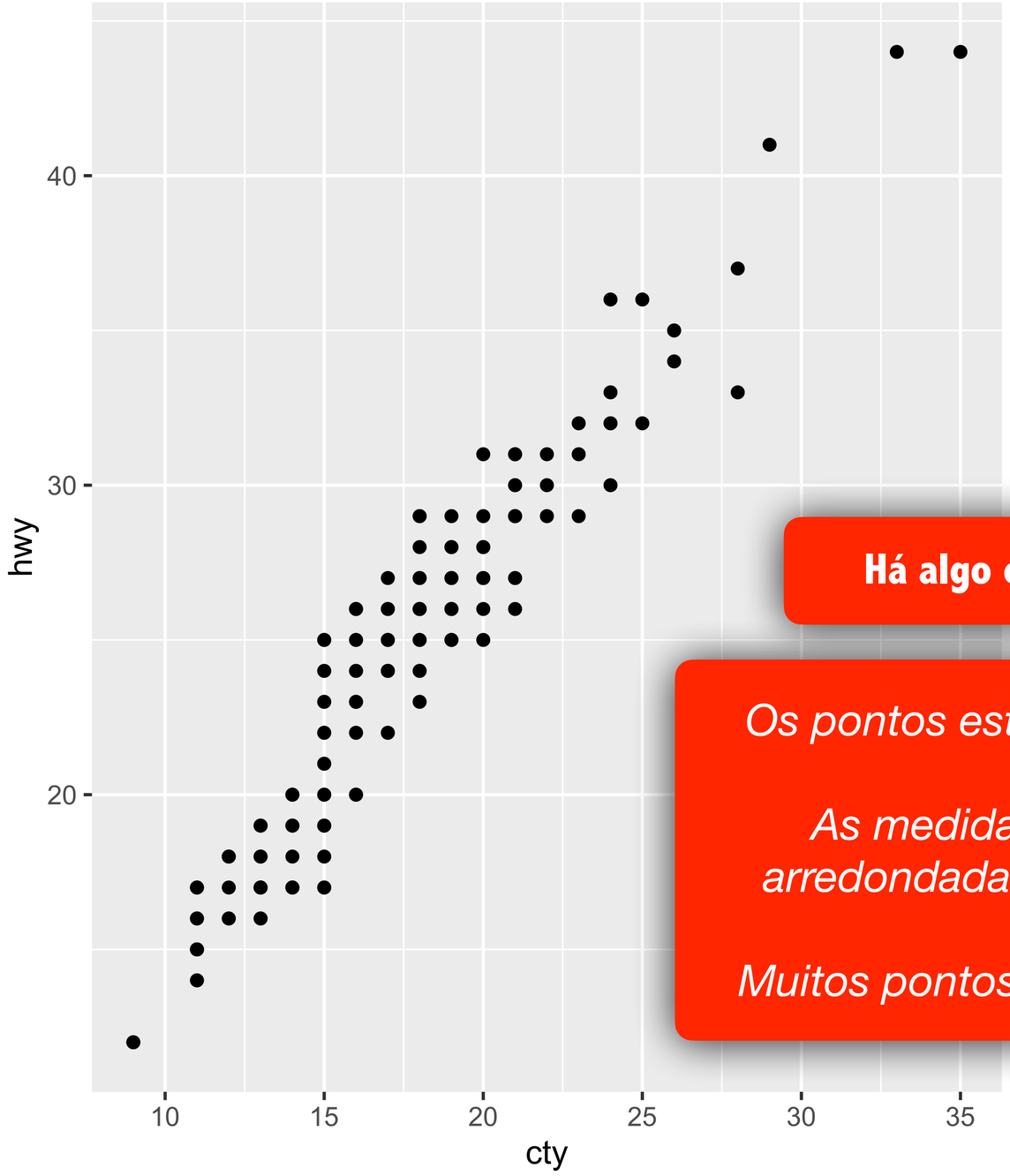

```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(fill = cut), position = "stack" )
```




```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(fill = cut), position = "fill" )
```




```
ggplot( diamonds, aes(x = color) ) +
  geom_bar( aes(fill = cut), position = "identity" )
```



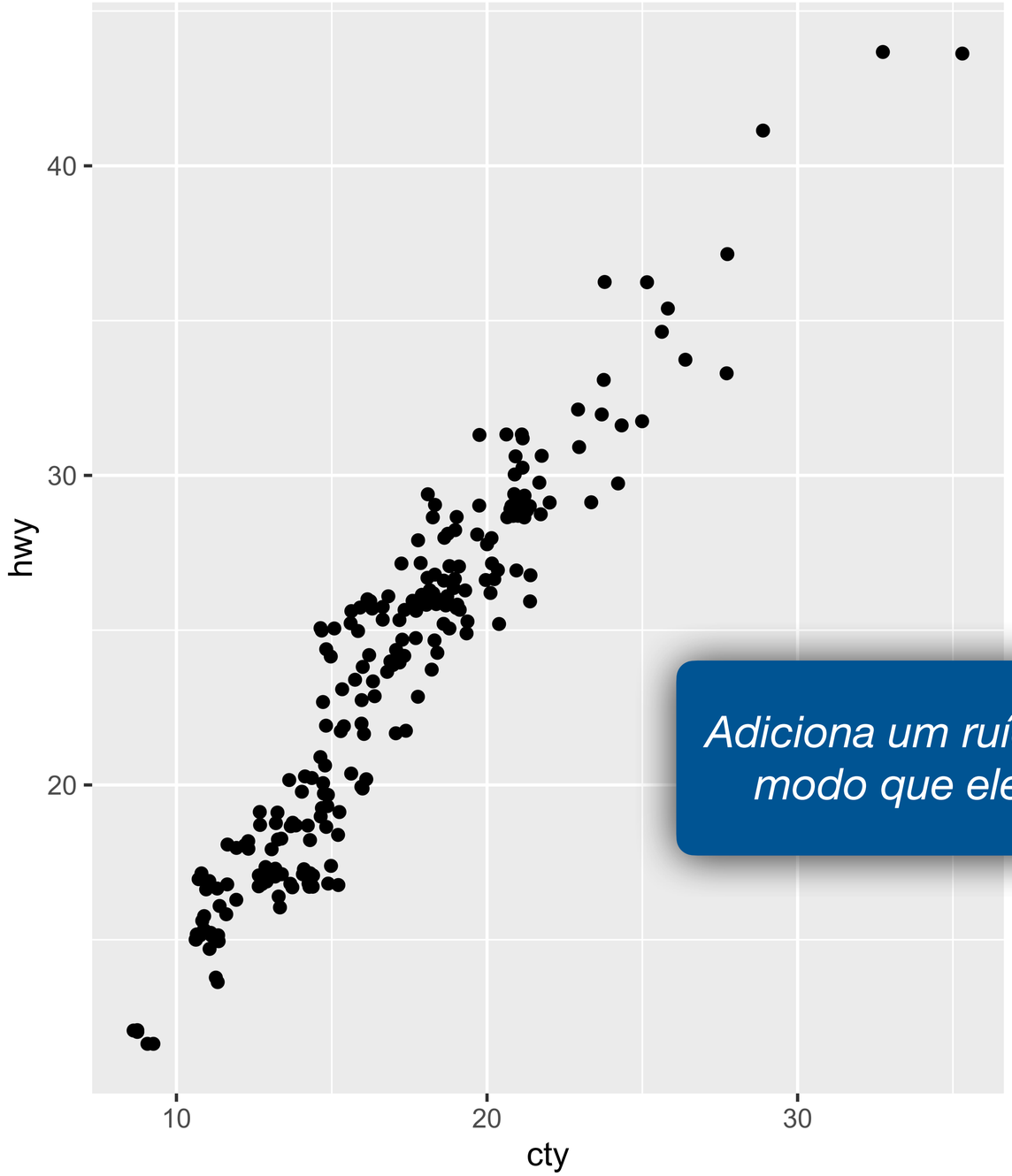

```
ggplot( mpg, aes(x = cty, y = hwy) ) +
  geom_point()
```

Há algo estranho nesse gráfico?

Os pontos estão arranjados em um grid.

As medidas provavelmente foram arredondadas ao inteiro mais próximo.

Muitos pontos devem estar sobrepostos.

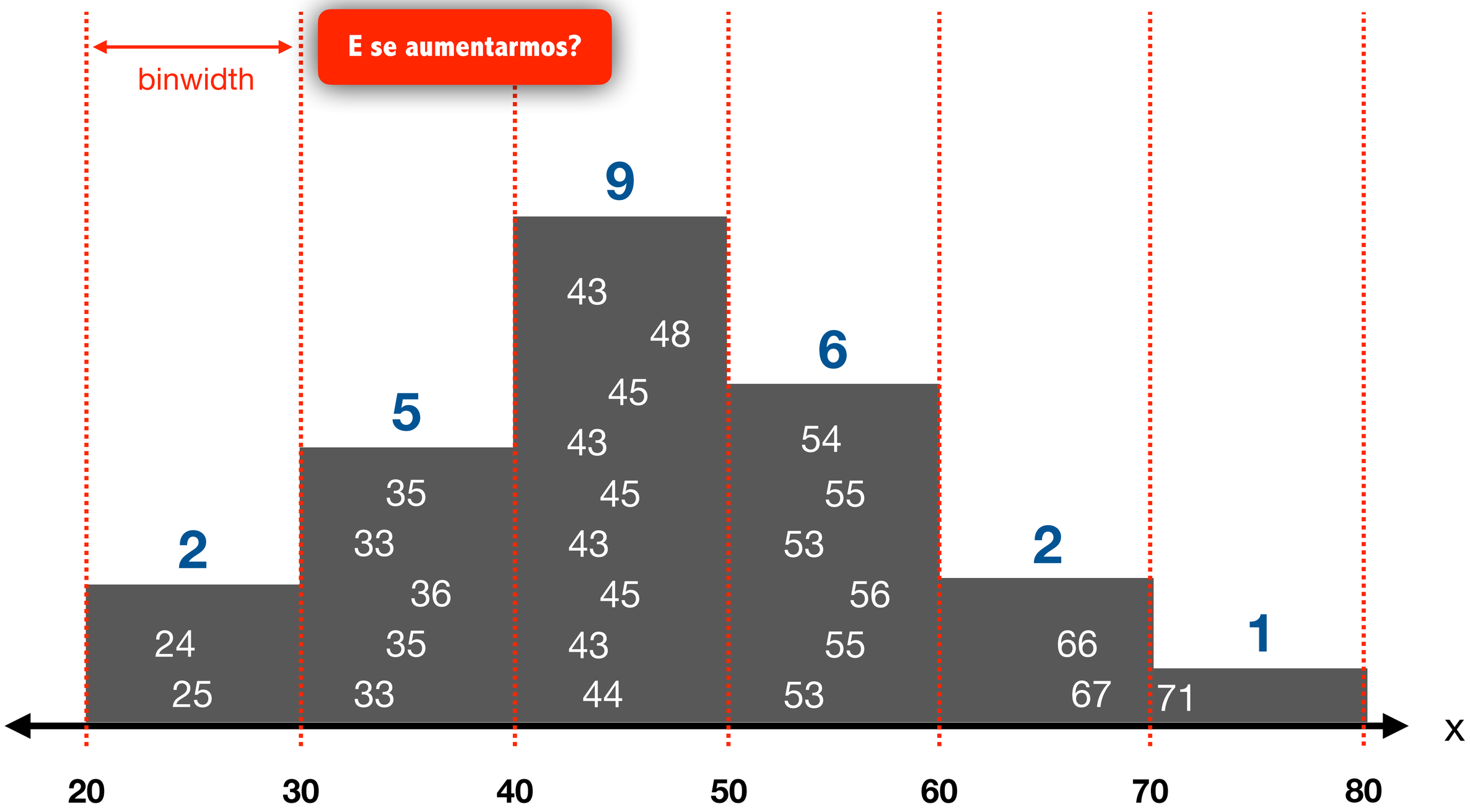


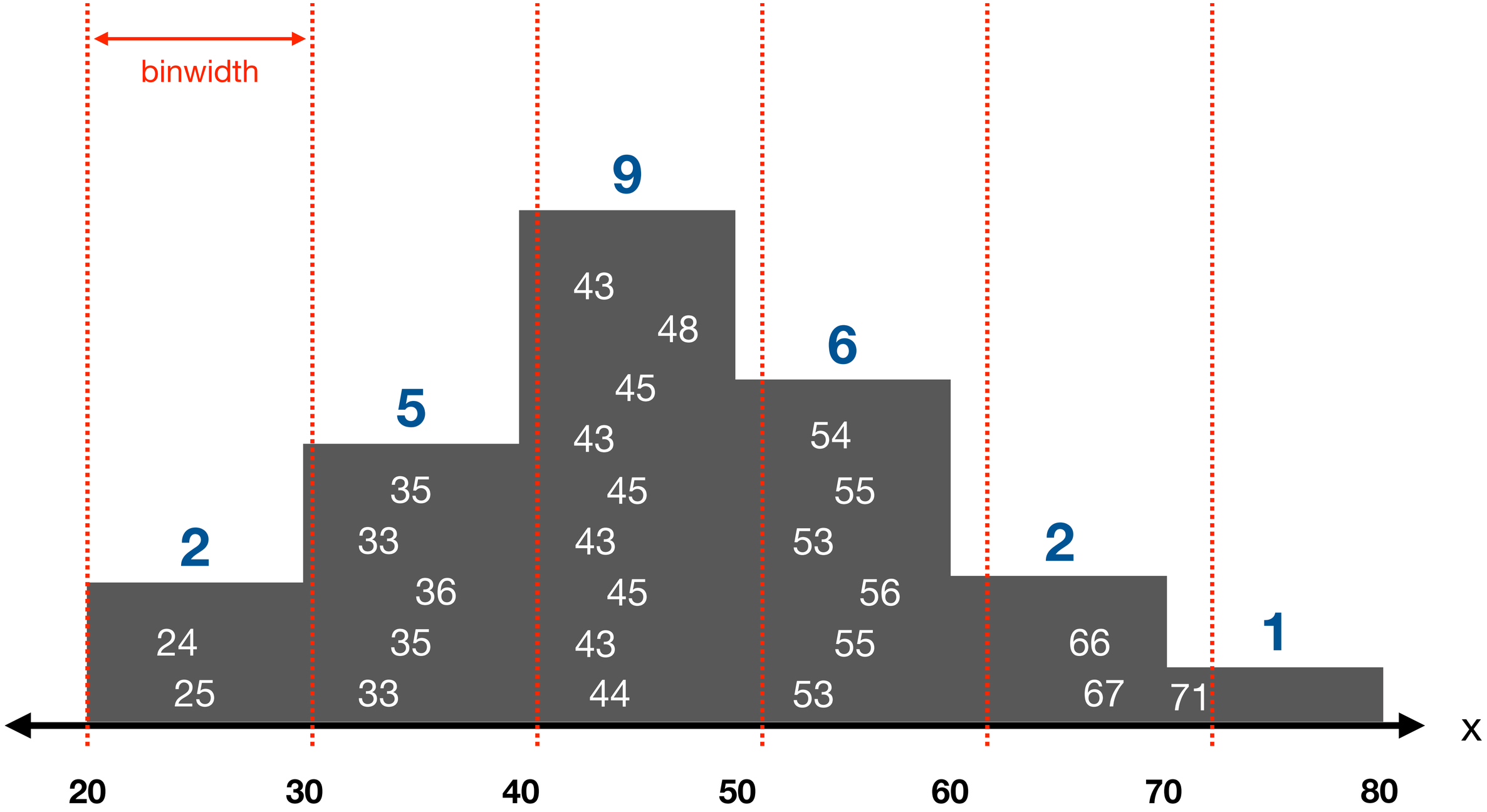

```
ggplot( mpg, aes(x = cty, y = hwy) ) +
  geom_jitter()
```

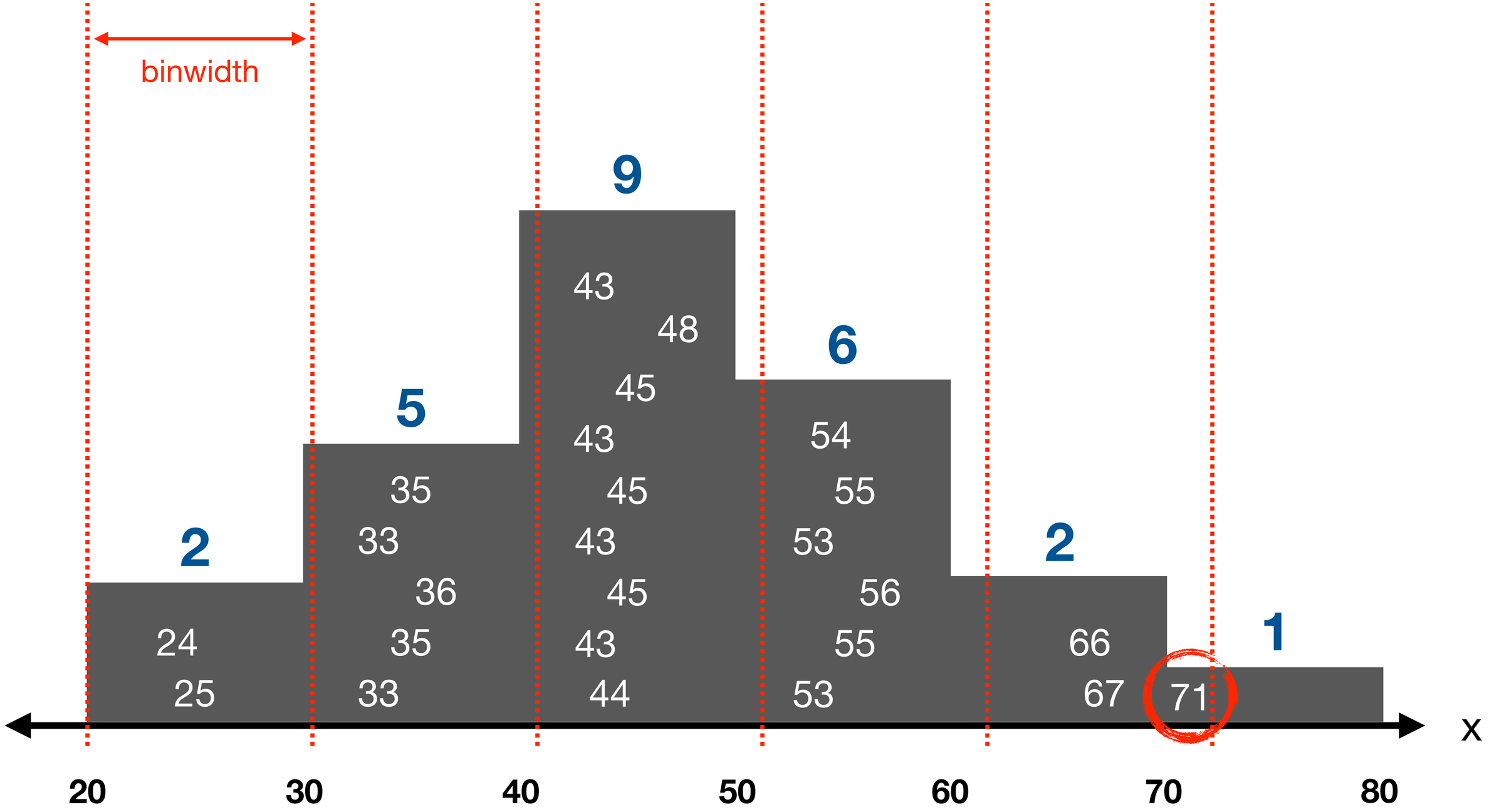
Adiciona um ruído aleatório a cada ponto de modo que eles passam a não sobrepor.

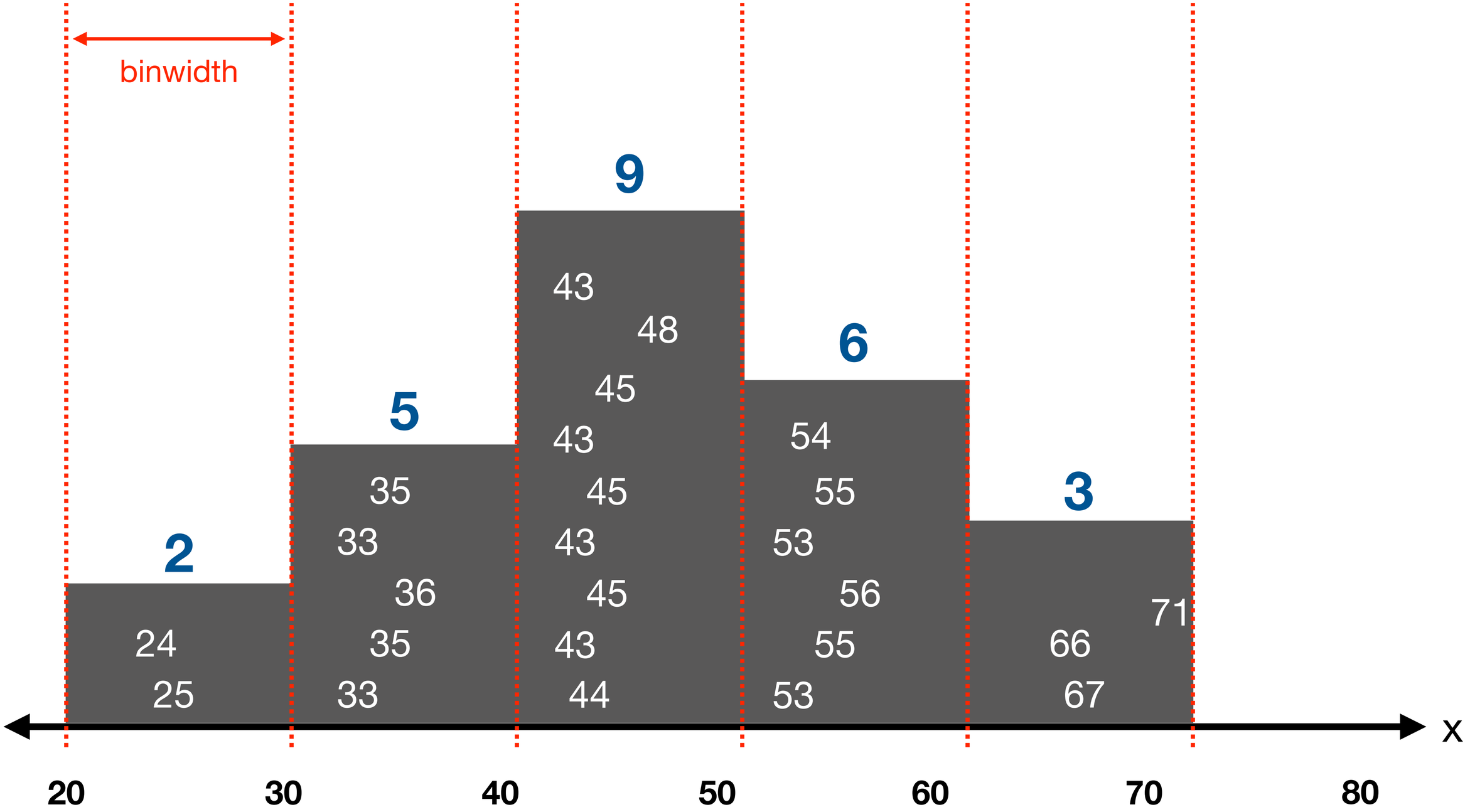
Histograma

1 dimensão: contínuo

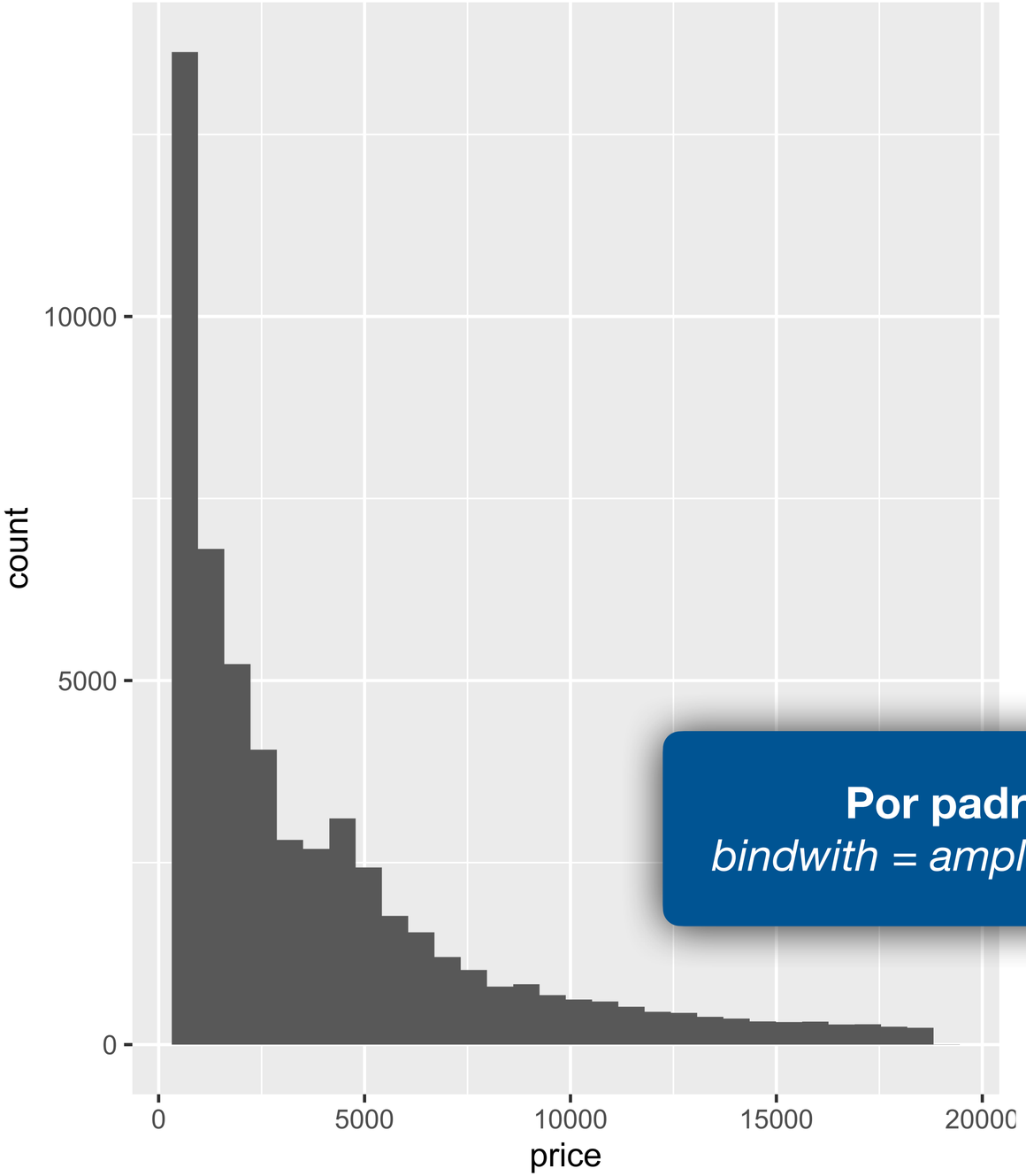

E se aumentarmos?
binwidth
2
5
9
6
2
1
24
25
35
33
36
35
33
43
48
45
43
45
43
45
44
54
55
53
56
55
53
66
67
71
x
20
30
40
50
60
70
80


binwidth
2
5
9
6
2
1
24
25
35
33
36
35
33
43
48
45
43
45
43
45
44
54
55
53
56
55
53
66
67
71
x
20
30
40
50
60
70
80


binwidth
2
5
9
6
2
1
24
25
35
33
36
35
33
43
48
45
43
45
43
45
44
54
55
53
56
55
53
66
67
71
x
20
30
40
50
60
70
80

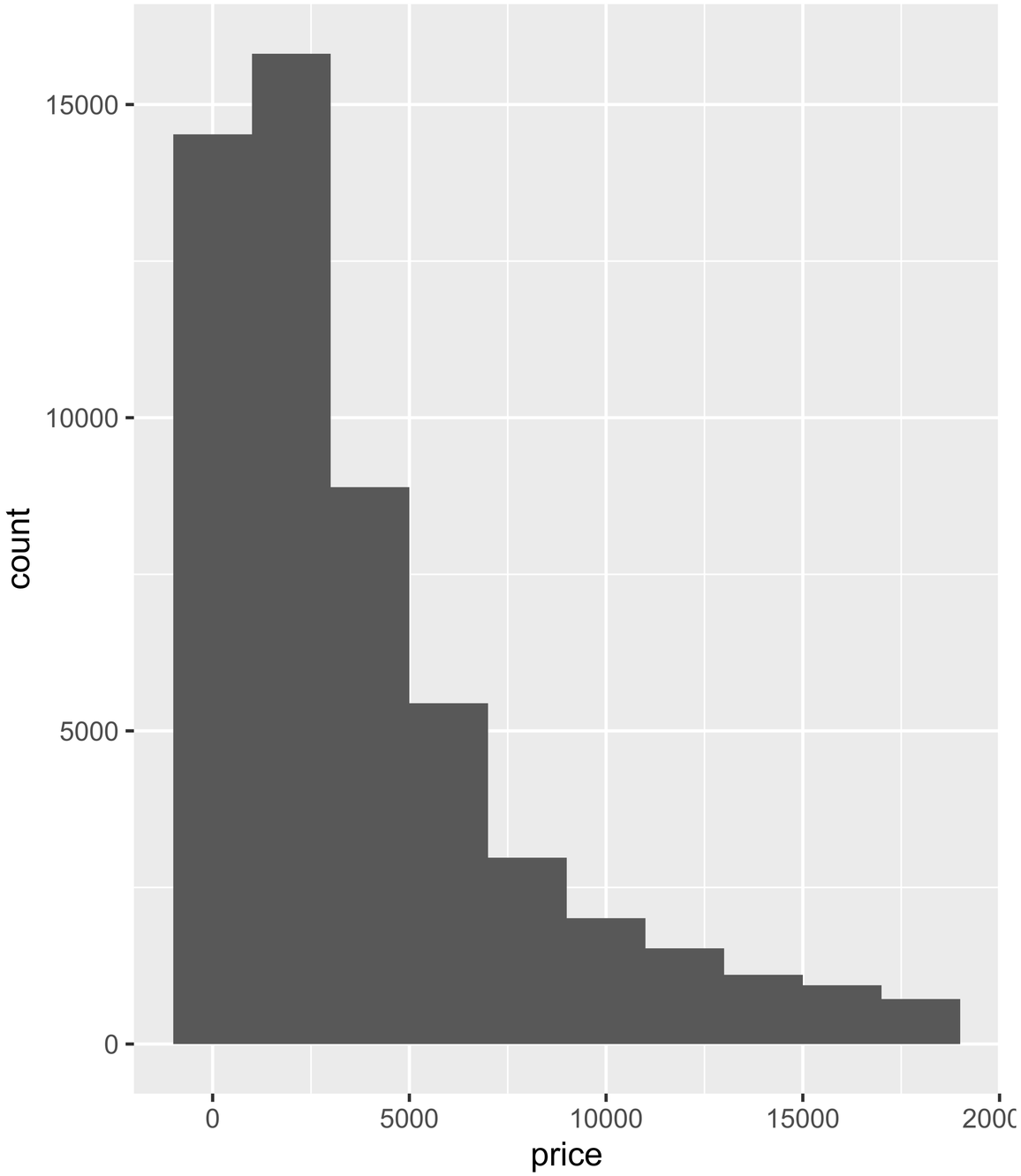

binwidth
2
24
25
5
35
33
36
35
33
9
43
48
45
43
45
43
45
43
44
6
54
55
53
56
55
53
3
71
66
67
x
20
30
40
50
60
70
80

Parâmetros



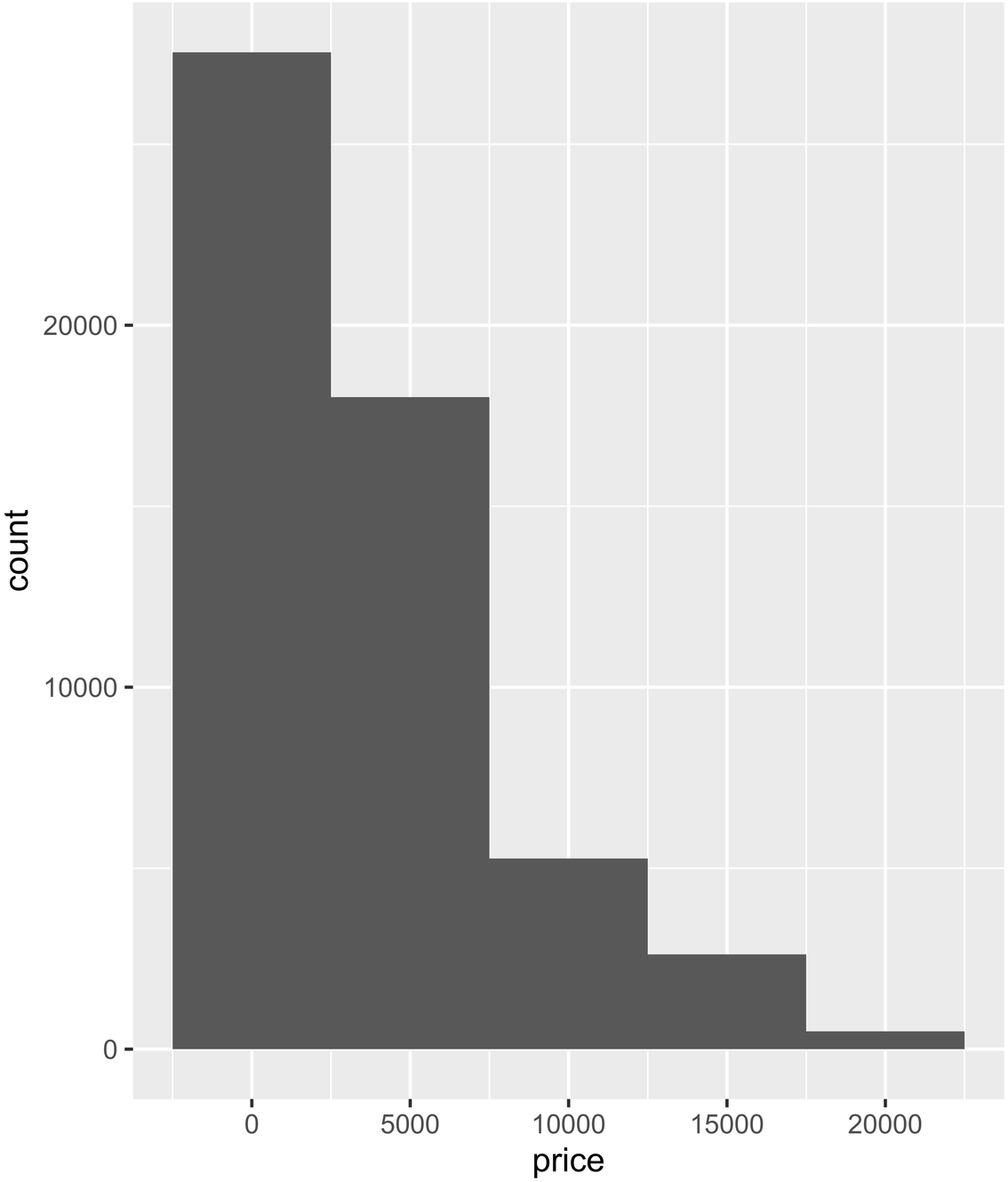

```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( )
```

Por padrão
bindwith = amplitude / 30

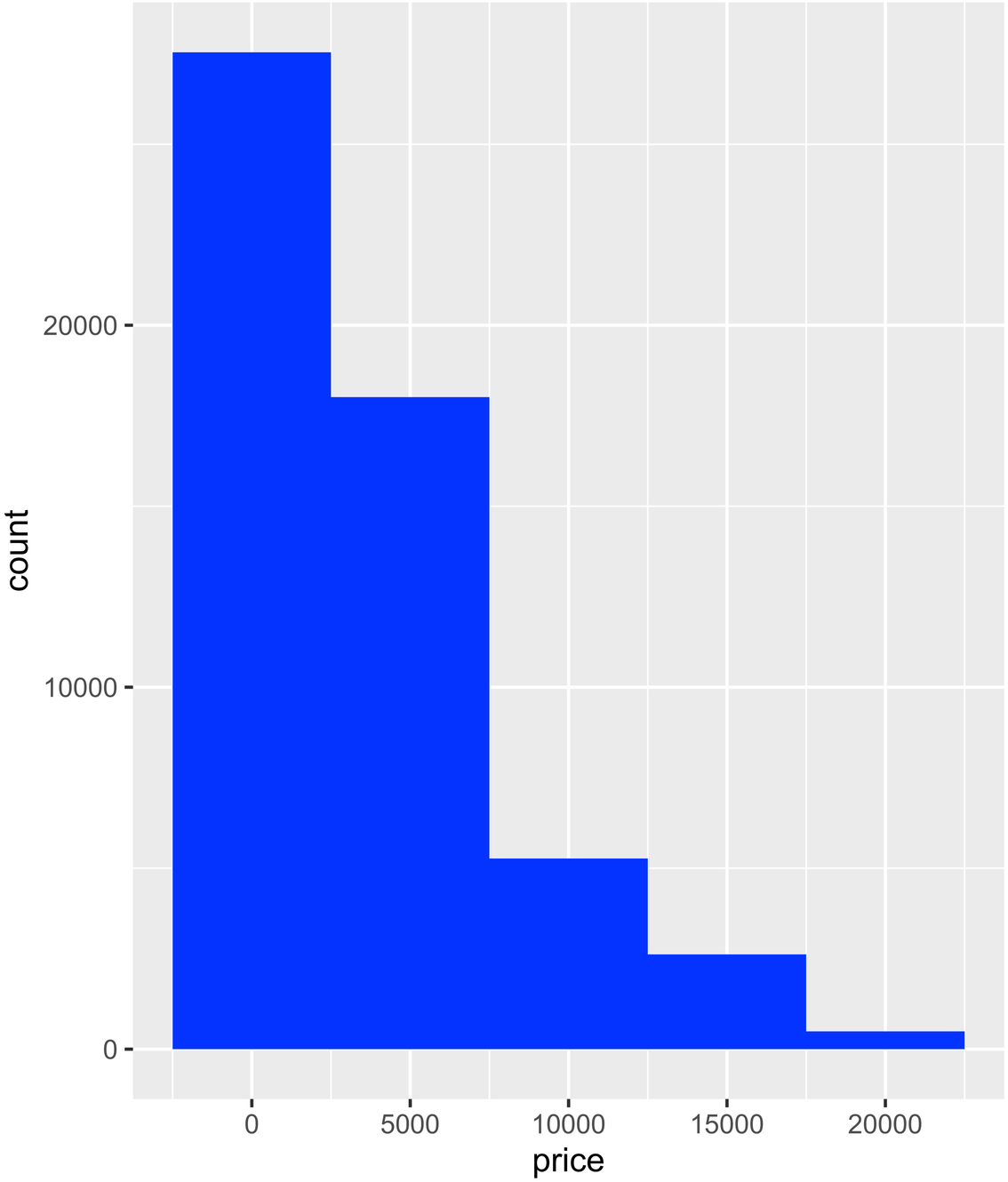



```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 2000 )
```

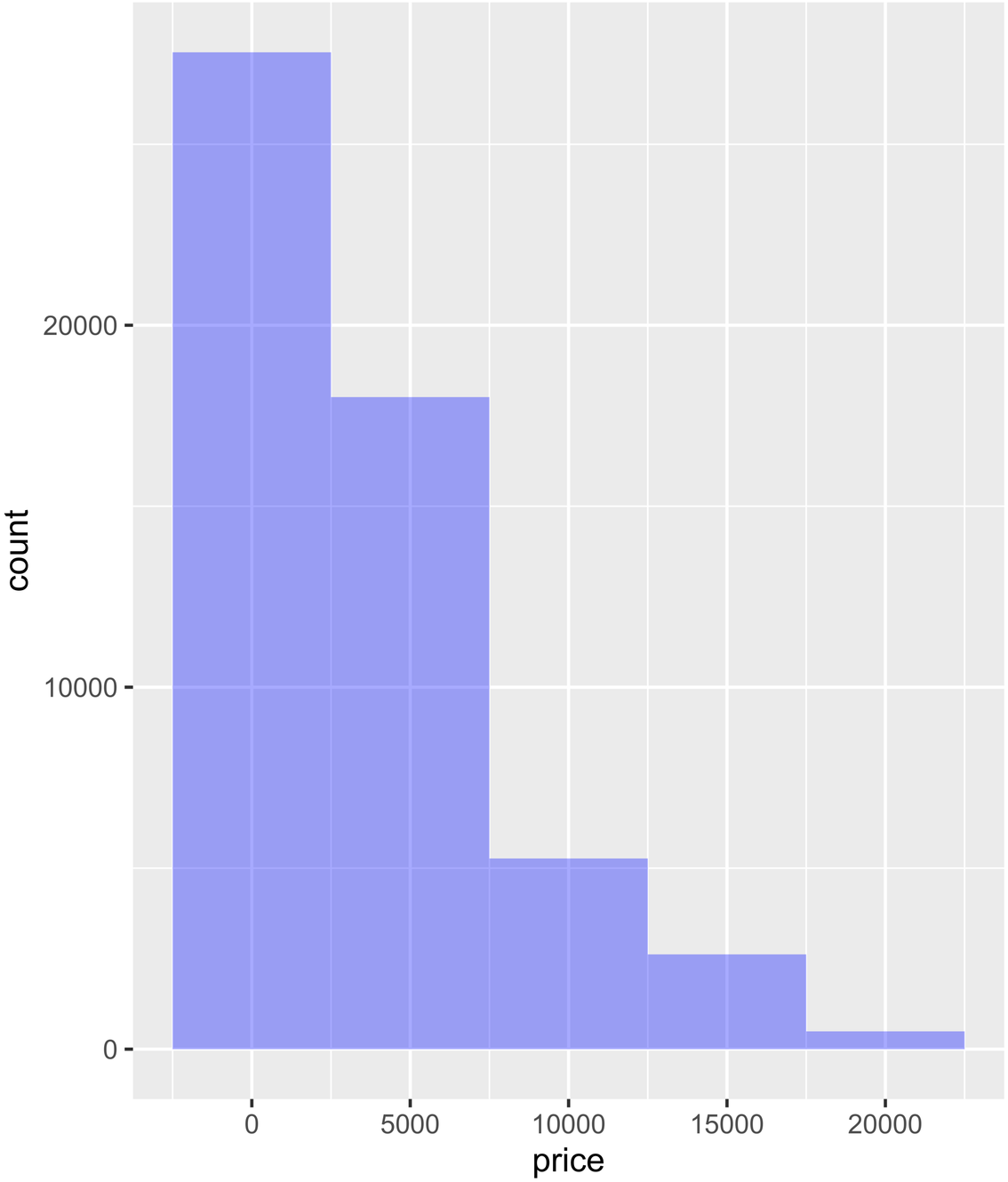
Parâmetro



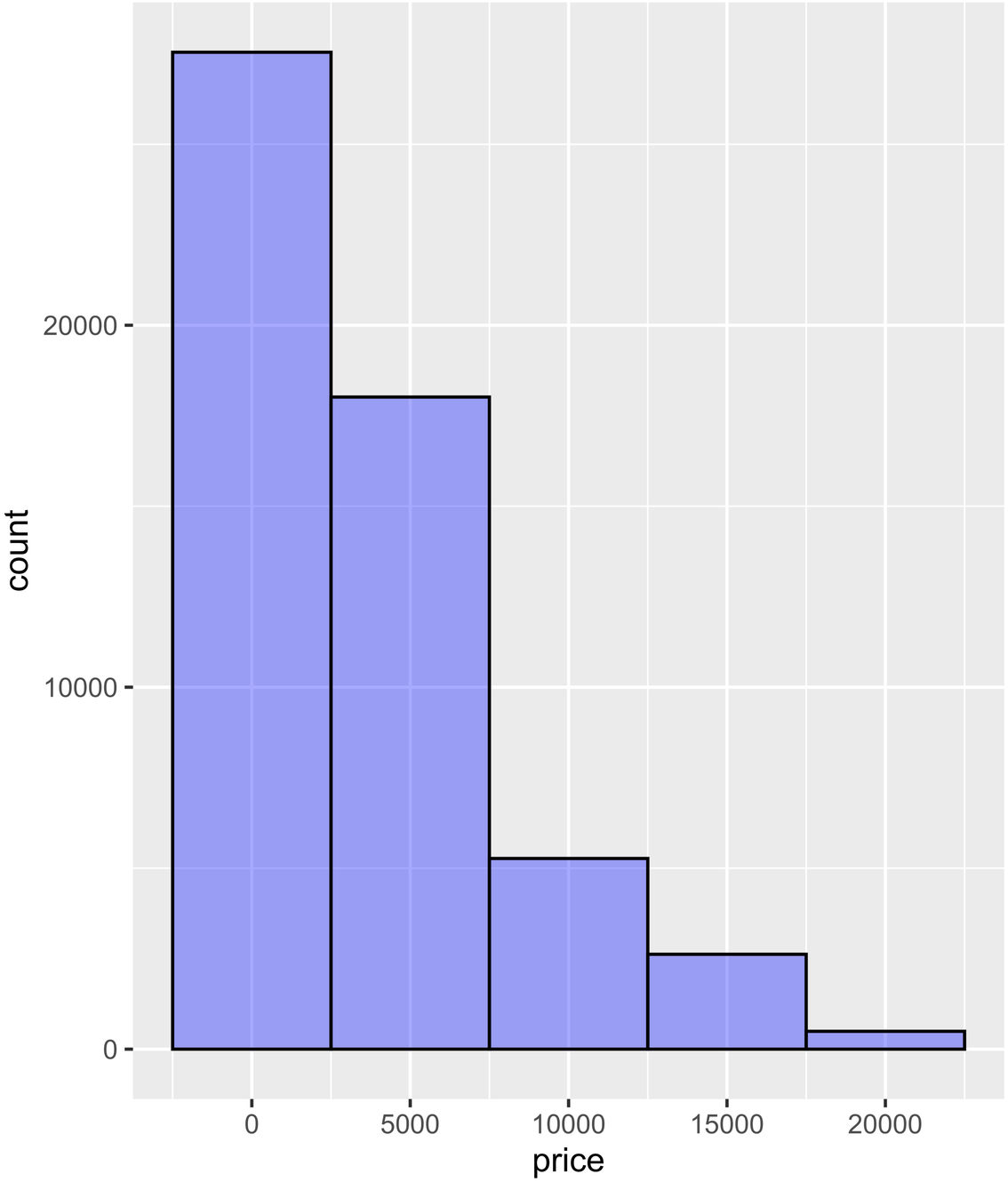

```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000 )
```



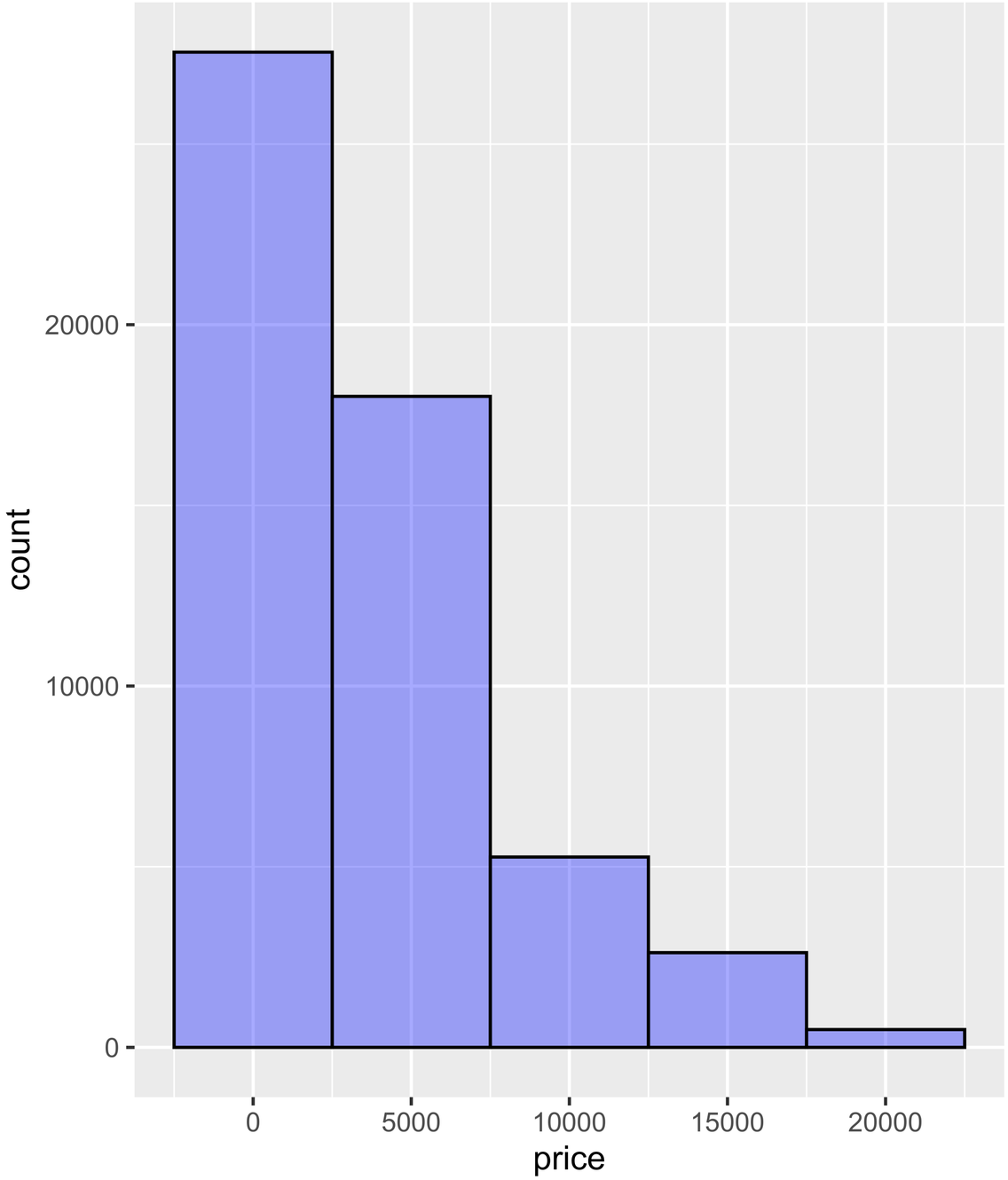

```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000,
                  fill = "blue" )
```




```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000,
                  fill = "blue",
                  alpha = 0.4 )
```

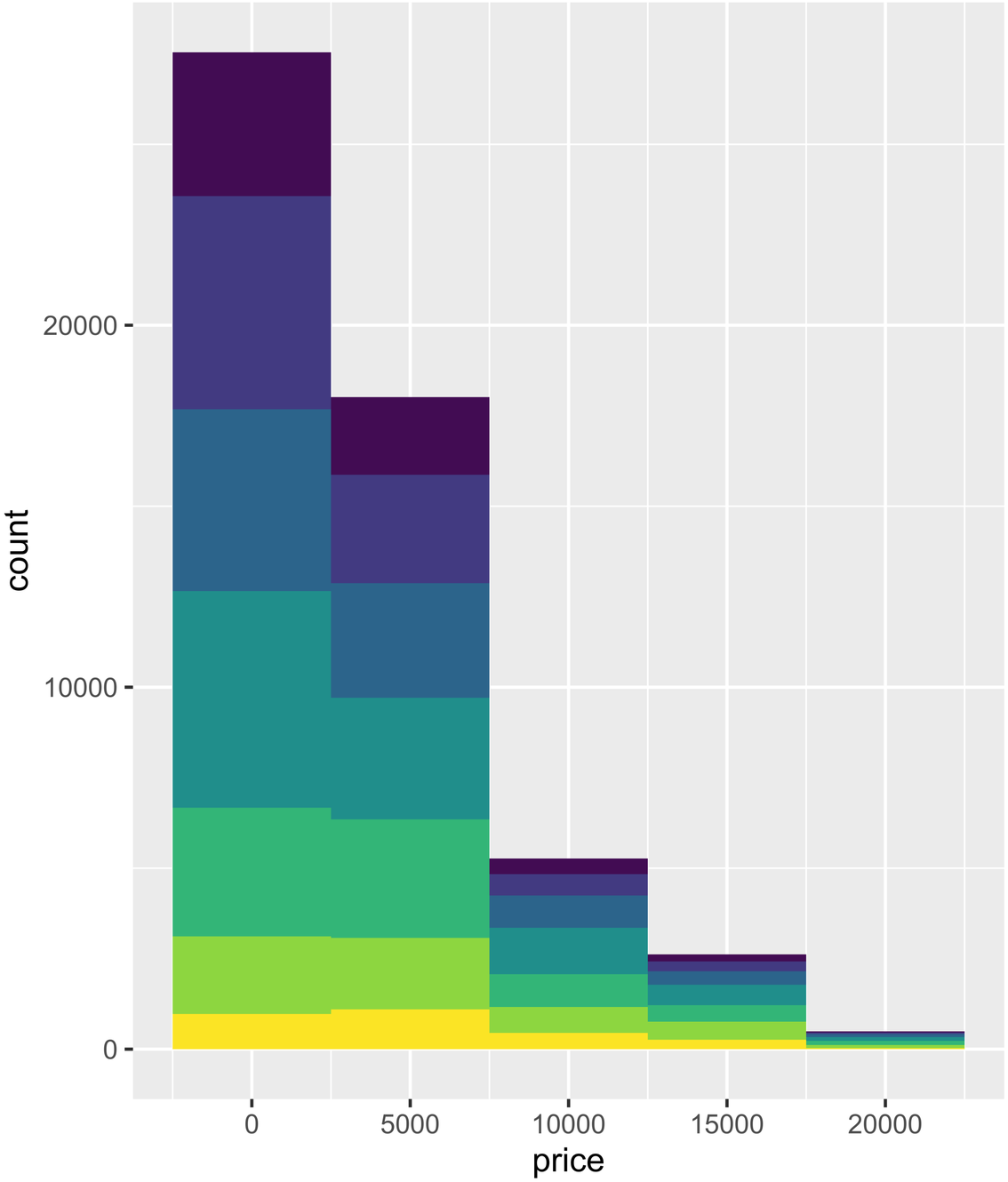



```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000,
                  fill = "blue",
                  alpha = 0.4,
                  color = "black" )
```

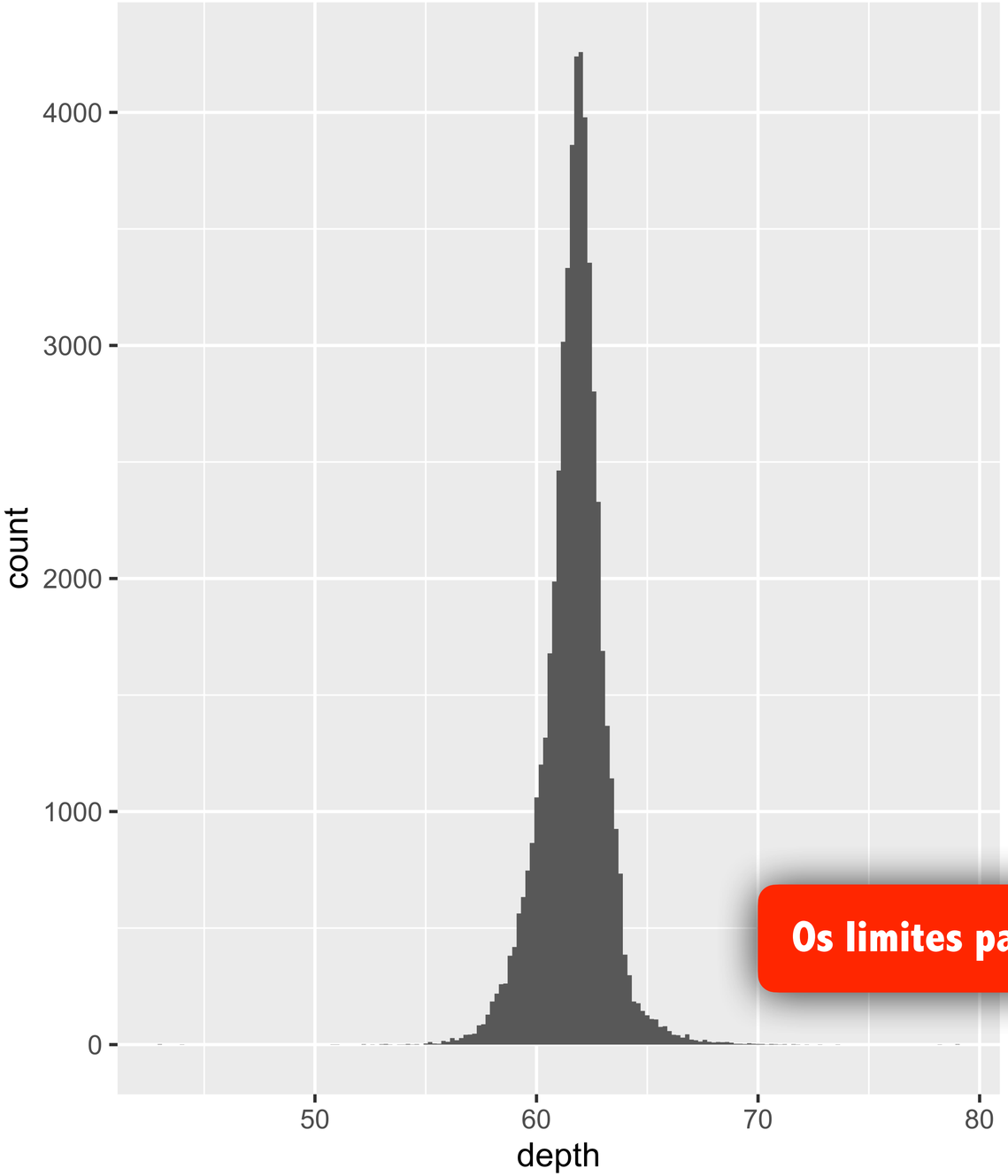



```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000,
                  fill = "blue",
                  alpha = 0.4,
                  color = "black" )
```

Controla a aparência do gráfico, mas não mapeia a aparência dos dados

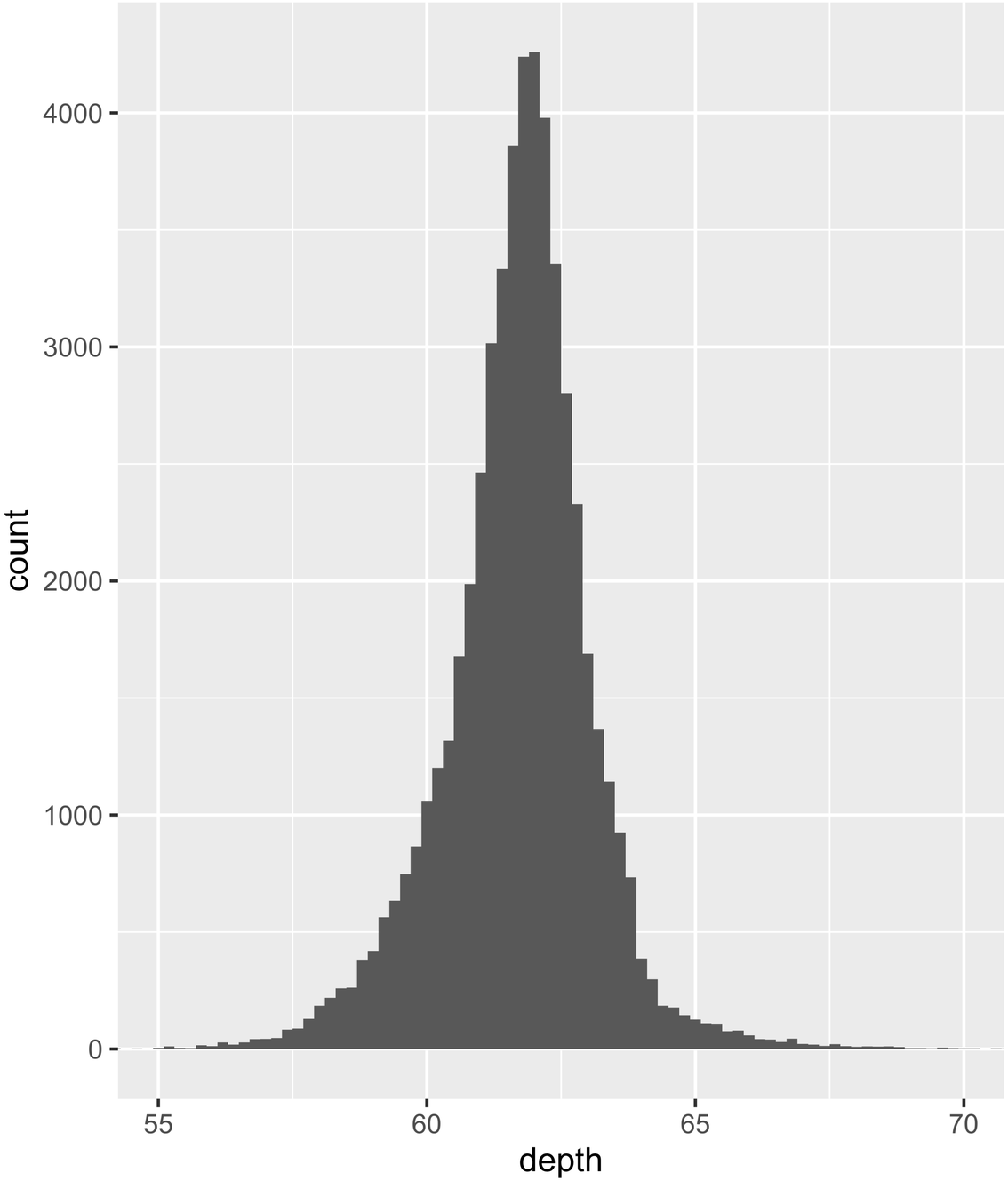



```
ggplot( diamonds, aes(x = price) ) +
  geom_histogram( binwidth = 5000,
                  aes( fill = color ),
                  show.legend = F )
```

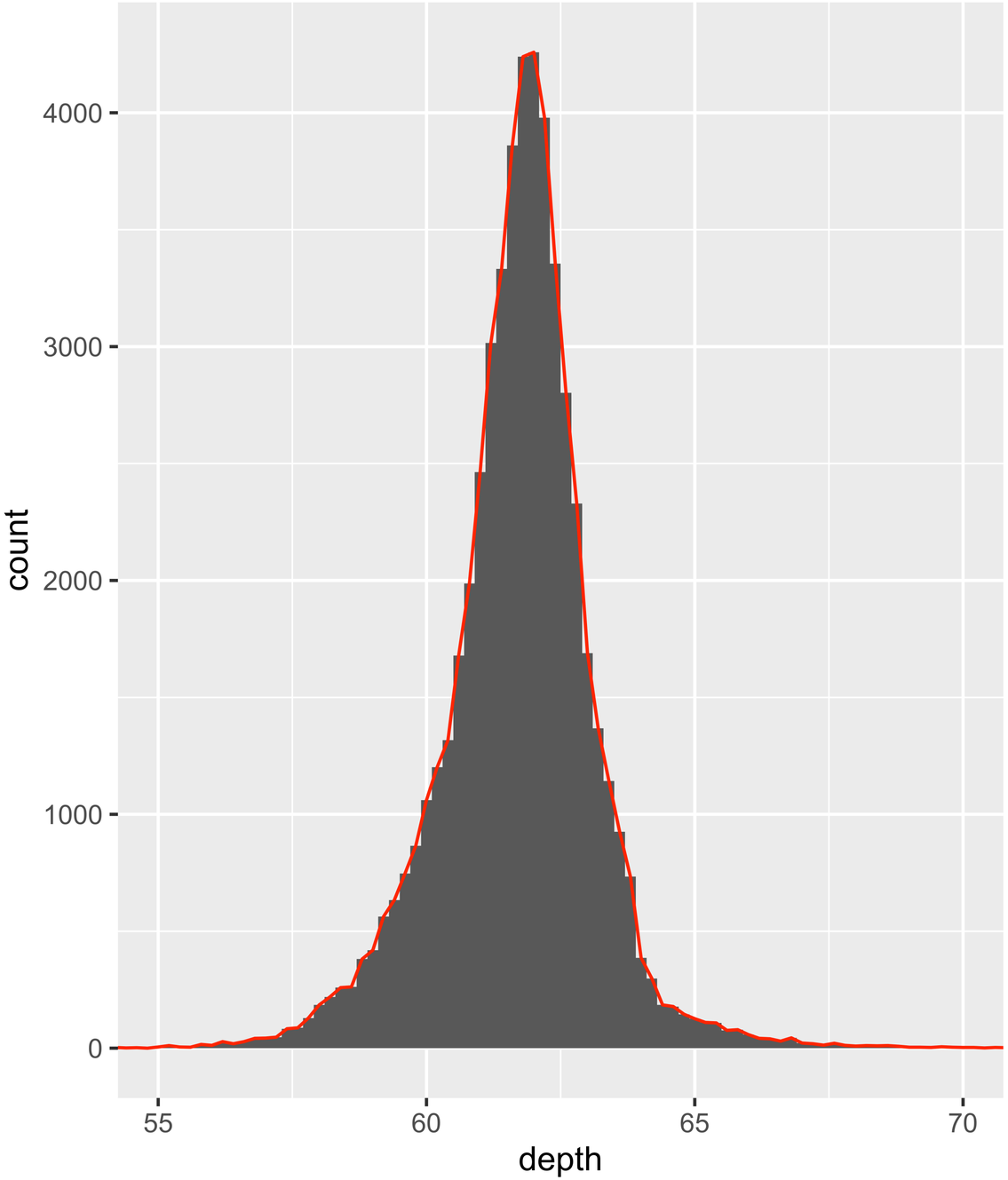



```
ggplot( diamonds, aes(x = depth) ) +
  geom_histogram( binwidth = 0.2 )
```

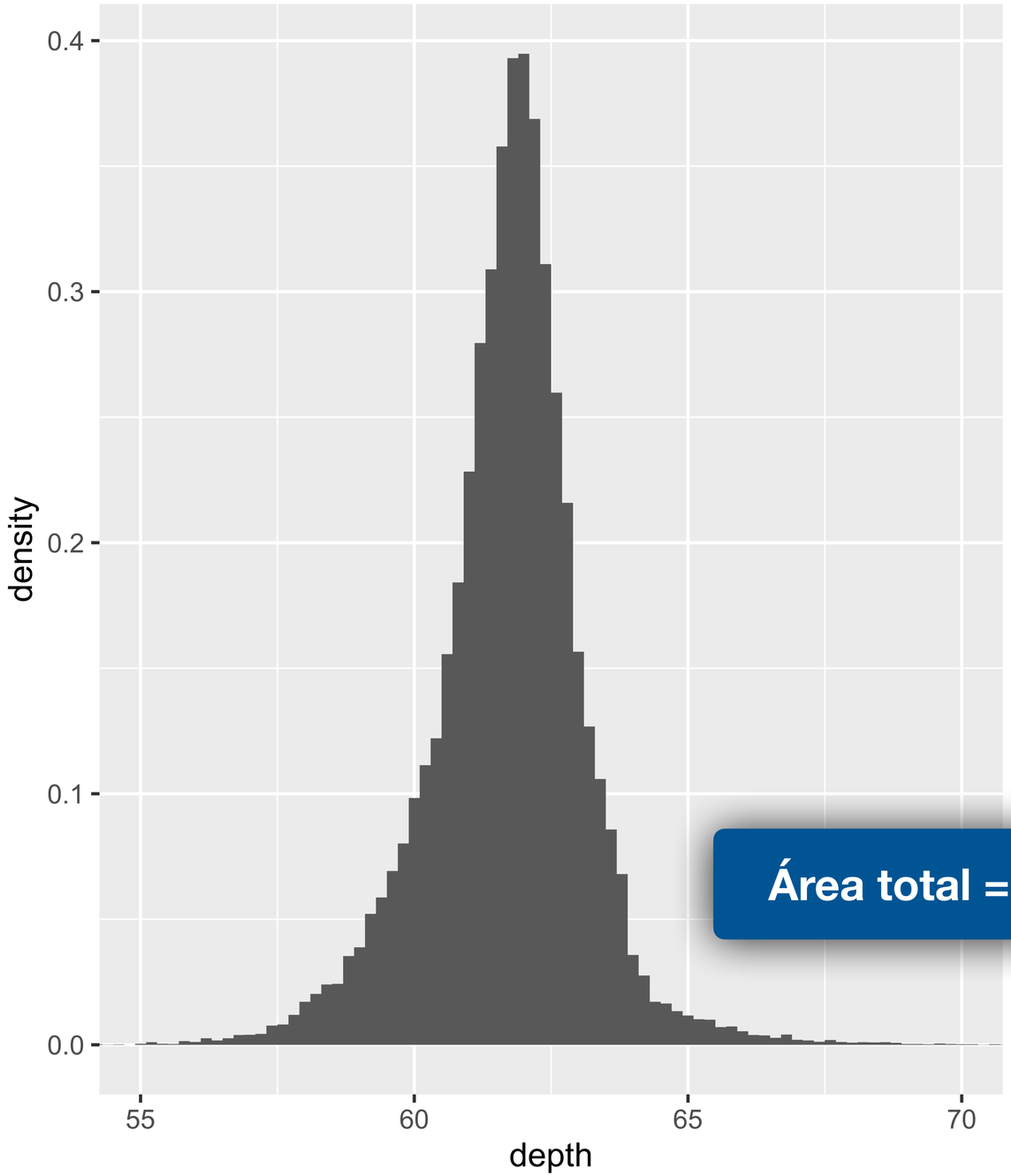
Os limites para a abcissa estão ruins!




```
ggplot( diamonds, aes(x = depth) ) +
  geom_histogram( binwidth = 0.2 ) +
  coord_cartesian( xlim = c(55,70) )
```

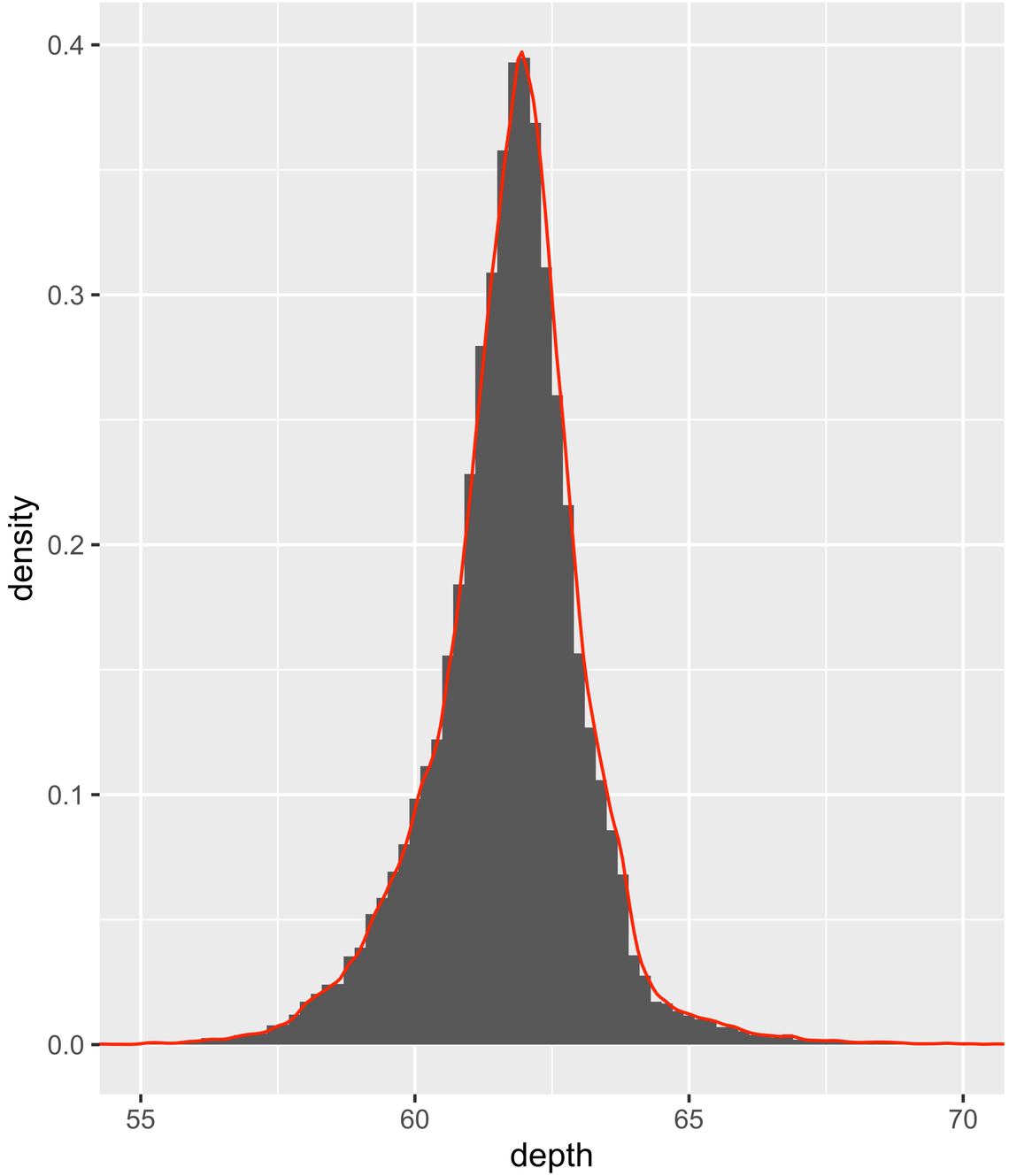



```
ggplot( diamonds, aes(x = depth) ) +
  geom_histogram( binwidth = 0.2 ) +
  coord_cartesian( xlim = c(55,70) ) +
  geom_freqpoly( binwidth = 0.2,
                 color = "red")
```



Área total = 1

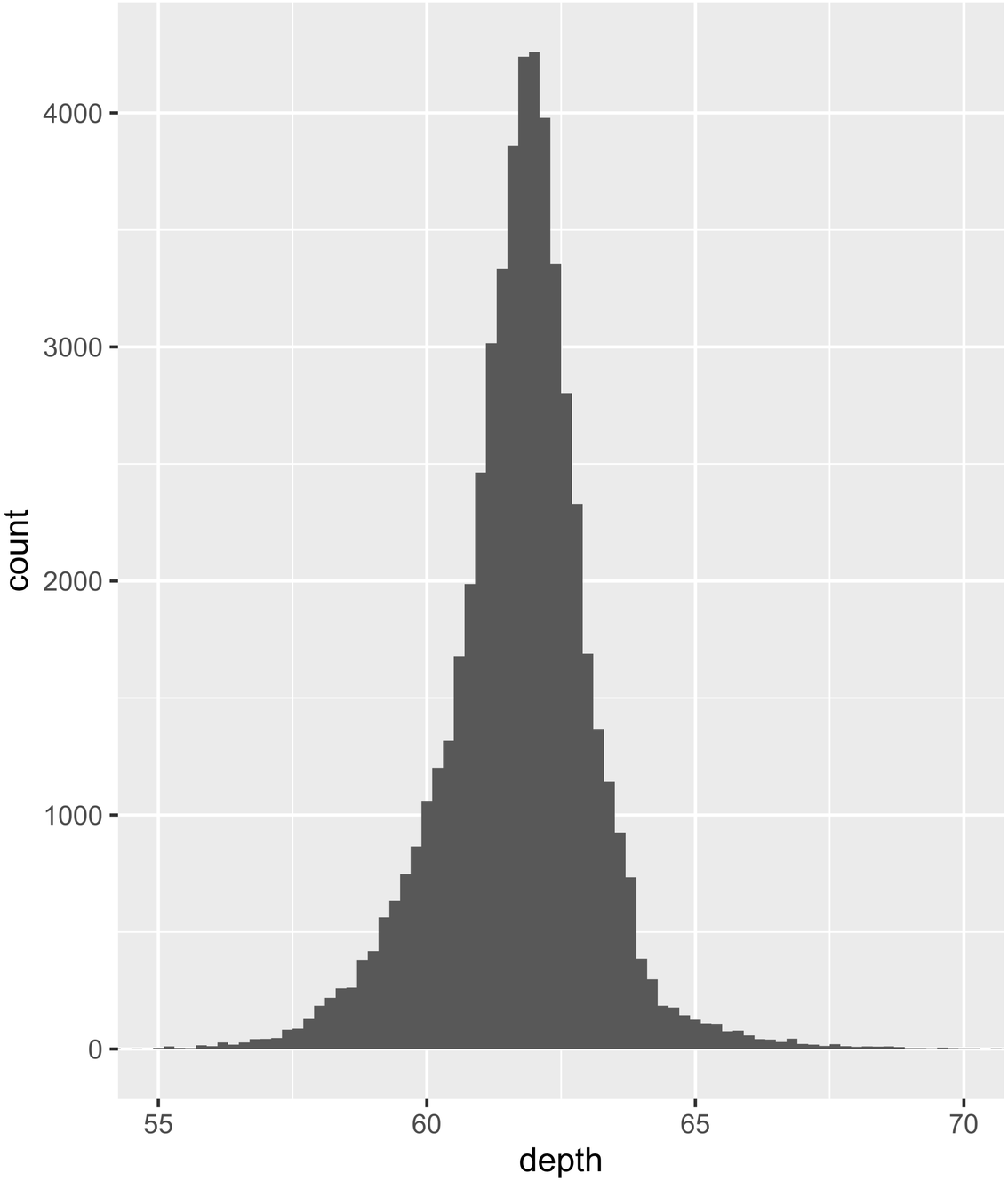
```
ggplot( diamonds, aes(x = depth) ) +
  geom_histogram(
    aes( y = after_stat(density) ),
         binwidth = 0.2 ) +
  coord_cartesian( xlim = c(55,70) )
```



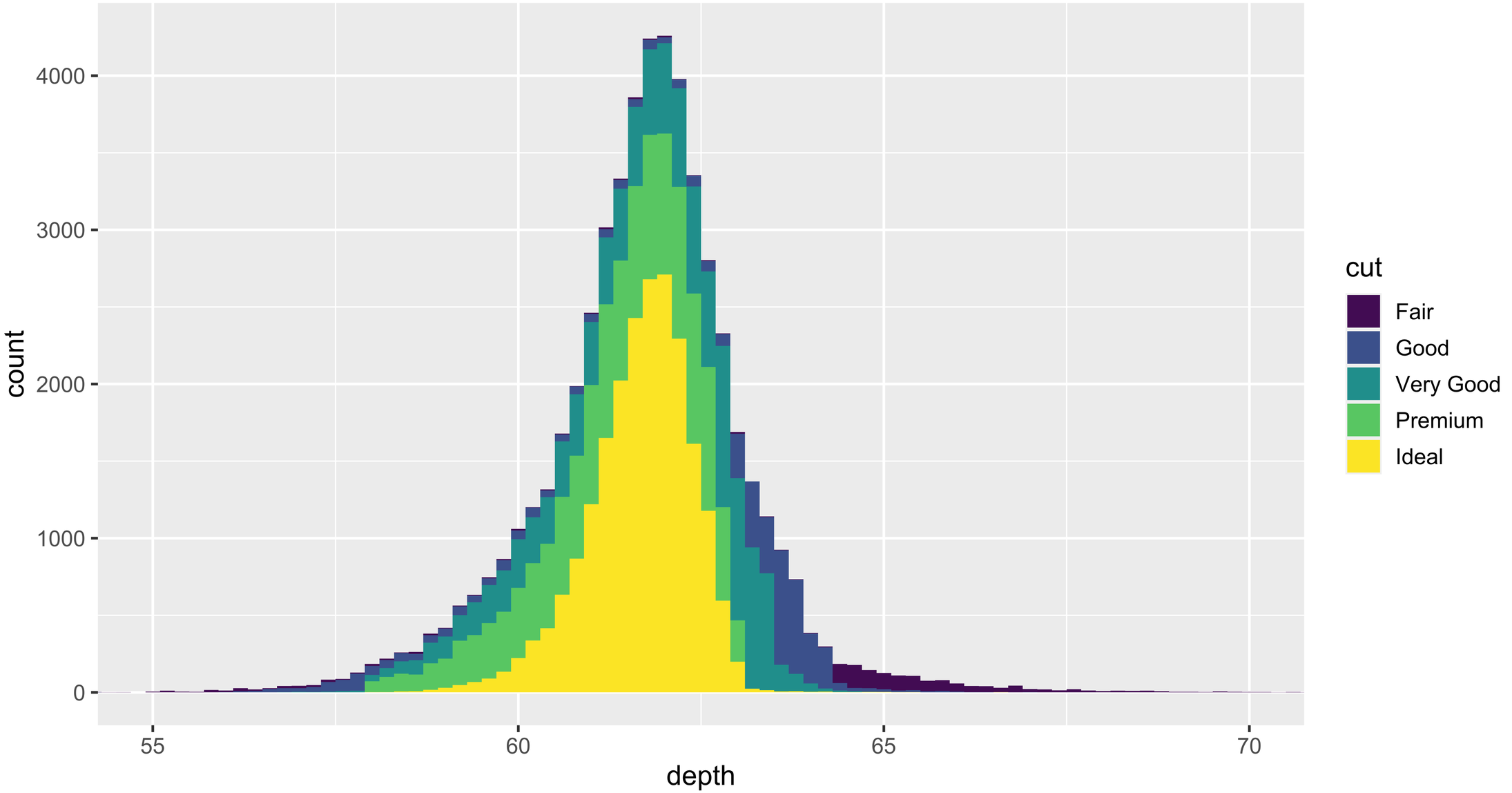

```
ggplot( diamonds, aes(x = depth) ) +
  geom_histogram(
    aes( y = after_stat(density) ),
         binwidth = 0.2 ) +
  coord_cartesian( xlim = c(55,70) ) +
  geom_density( color = "red" )
```

Você pode salvar elementos do ggplot, em objetos, para uso posterior

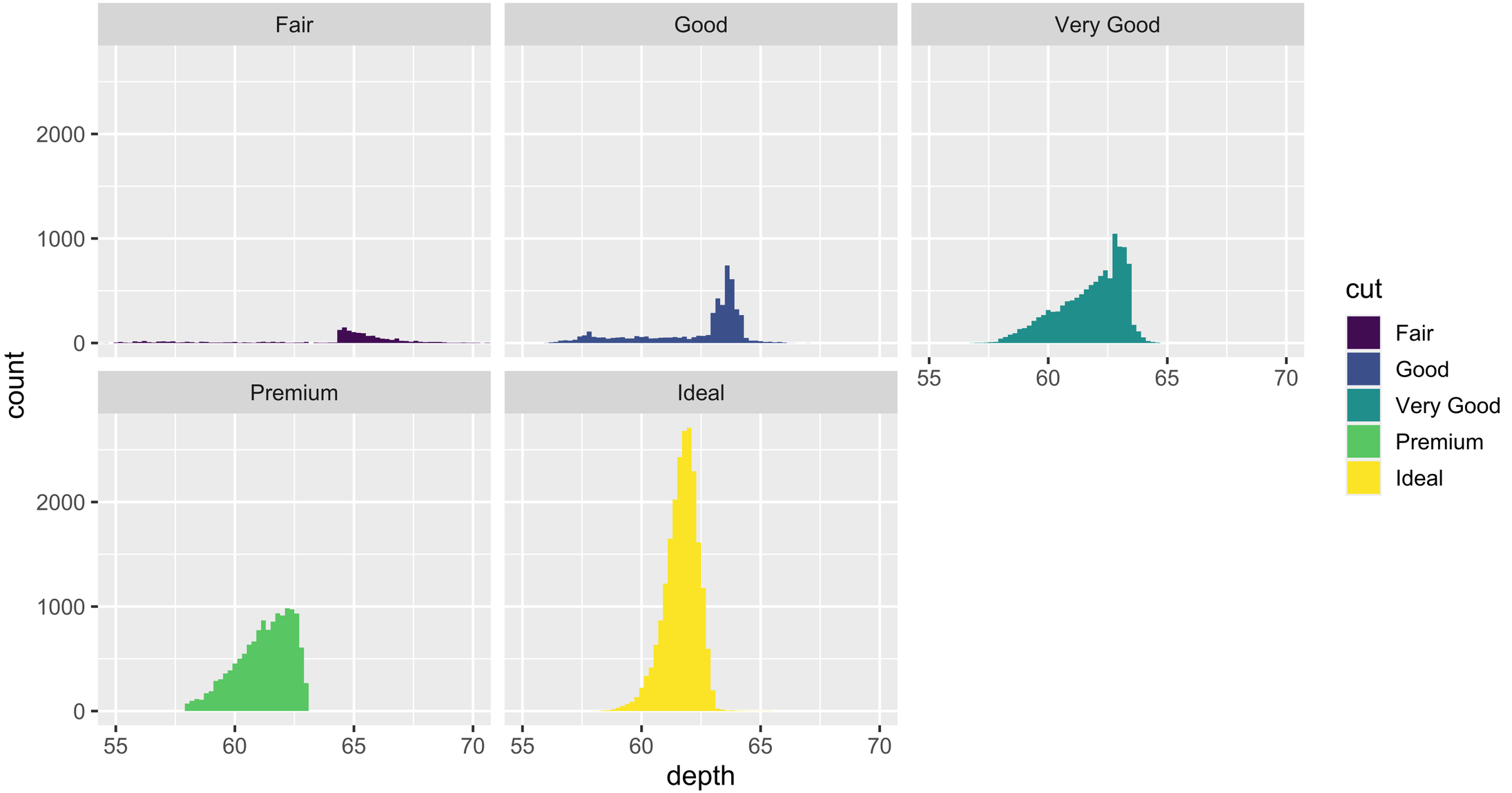
```
d_depth <- ggplot( diamonds, aes(x = depth) ) +
             coord_cartesian( xlim = c(55,70) )
```



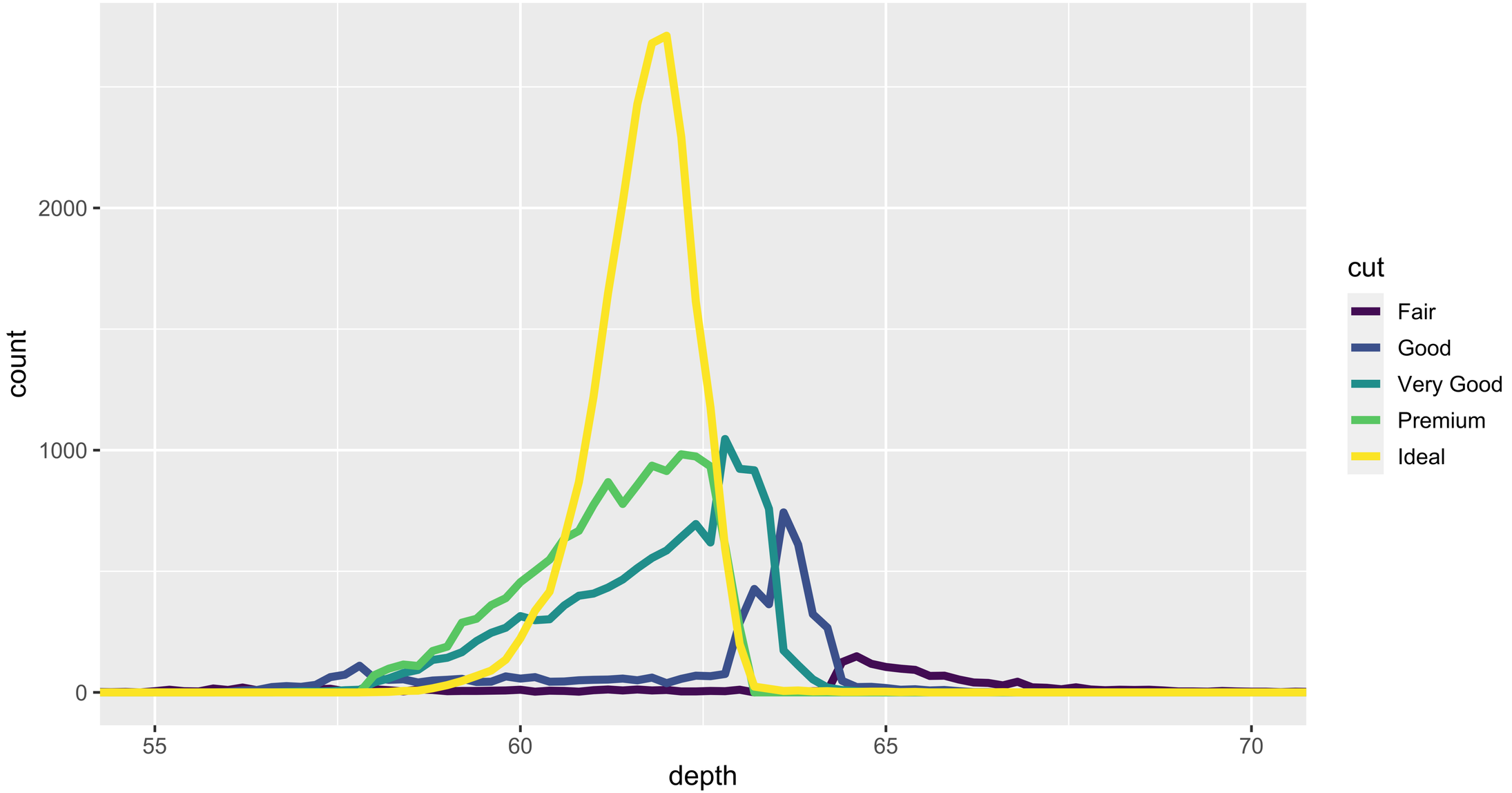

```
d_depht +
  geom_histogram( binwidth = 0.2 )
```



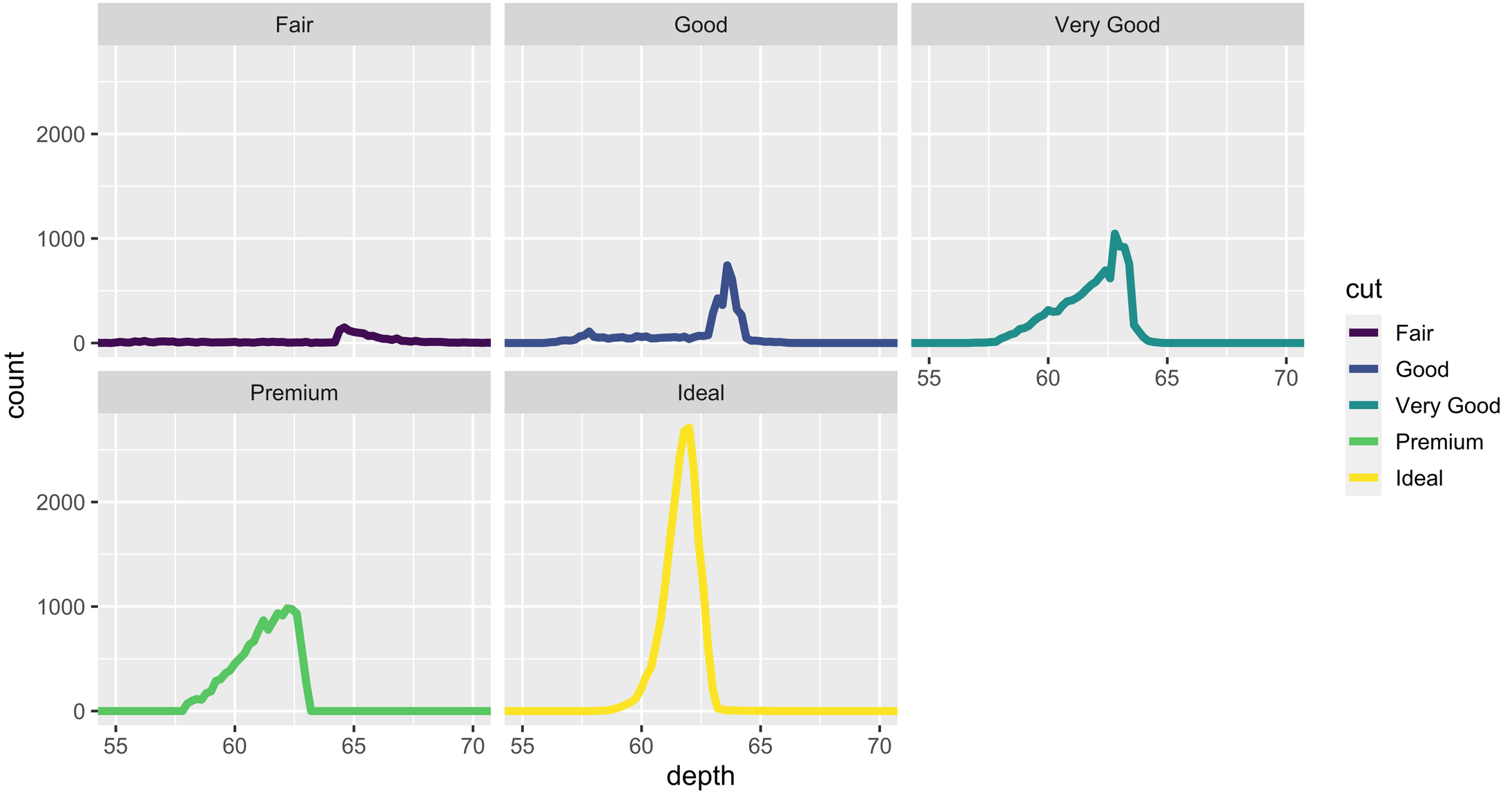

```
d_depht + geom_histogram( binwidth = 0.2, aes(fill = cut) )
```



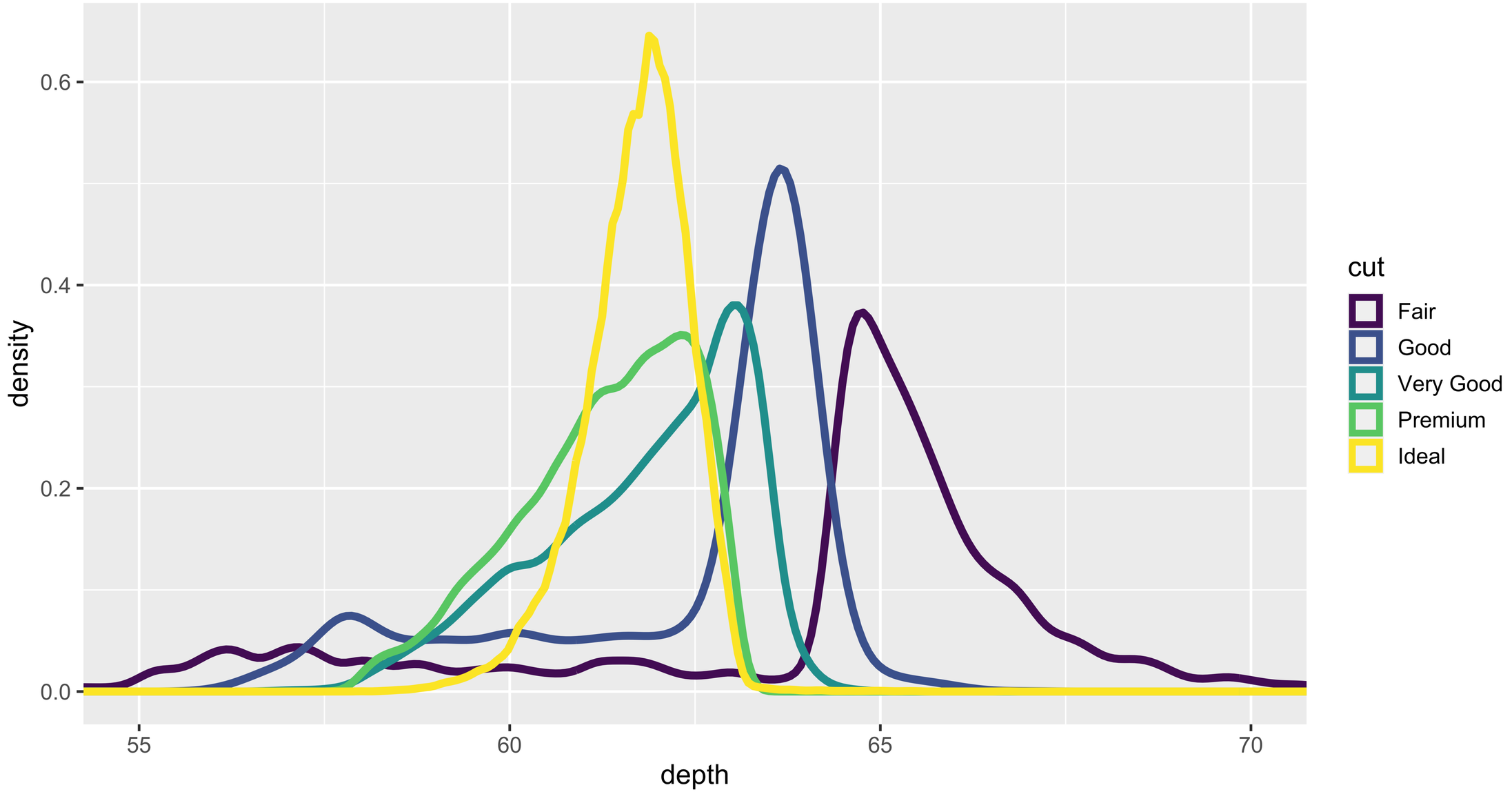

```
d_depht + geom_histogram( binwidth = 0.2, aes(fill = cut) ) +
  facet_wrap( ~ cut)
```



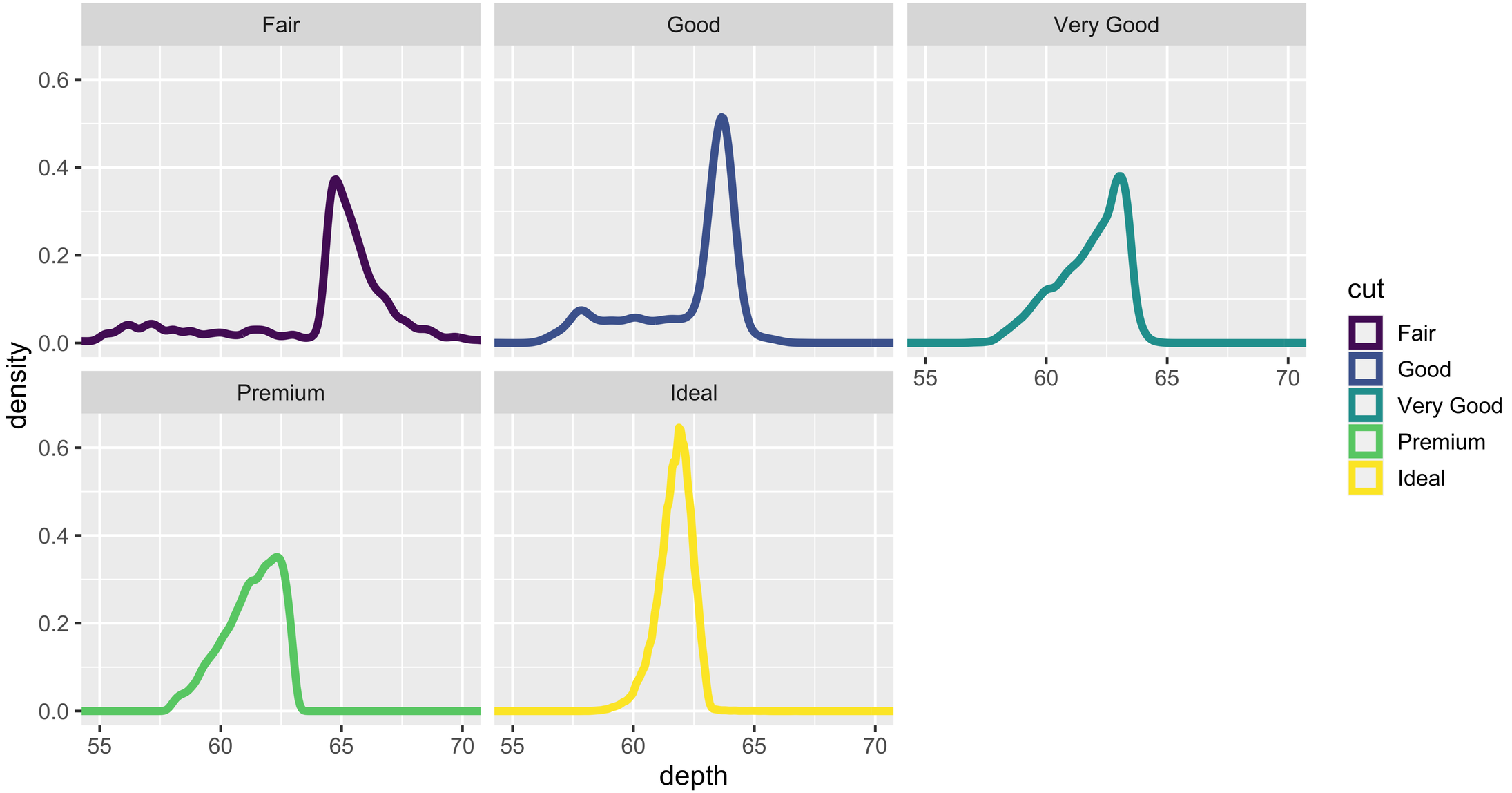

```
d_depht + geom_freqpoly( binwidth = 0.2, aes(color = cut) )
```




```
d_depht + geom_freqpoly( binwidth = 0.2, aes(color = cut) ) +
  facet_wrap( ~ cut)
```



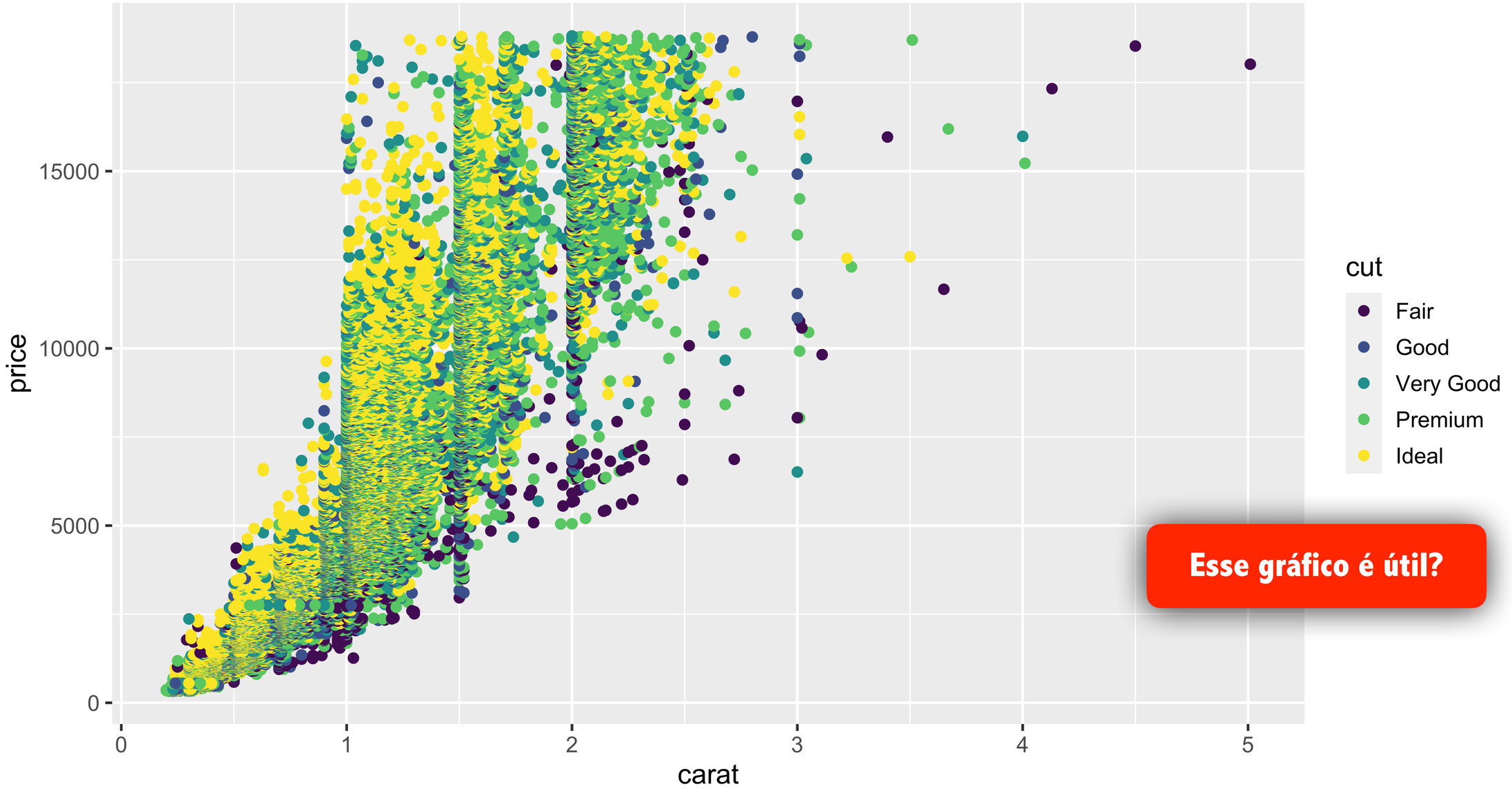

```
d_depht + geom_density( aes(color = cut) )
```



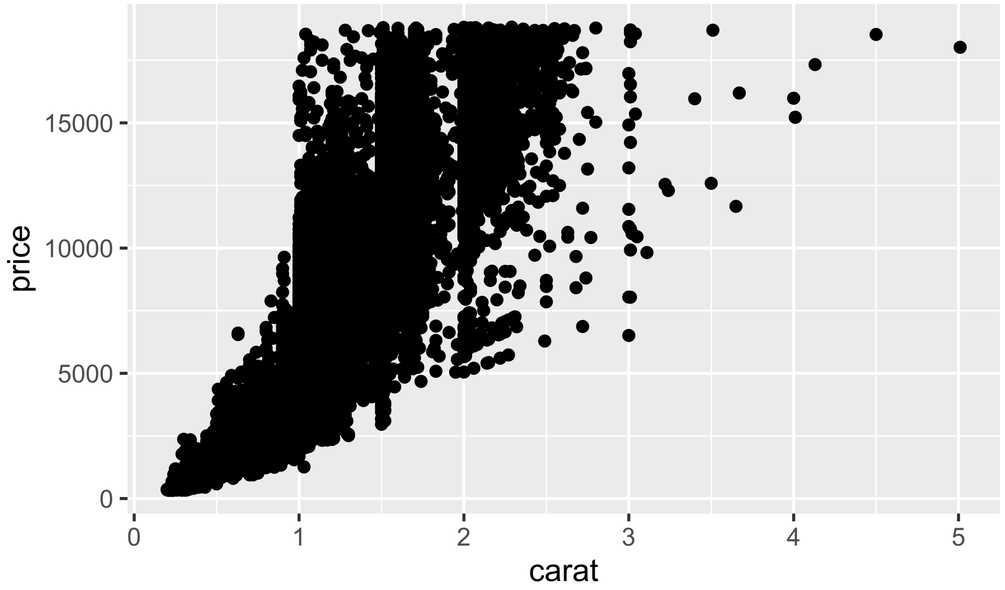

```
d_depht + geom_density( aes(color = cut) ) +
  facet_wrap( ~ cut)
```

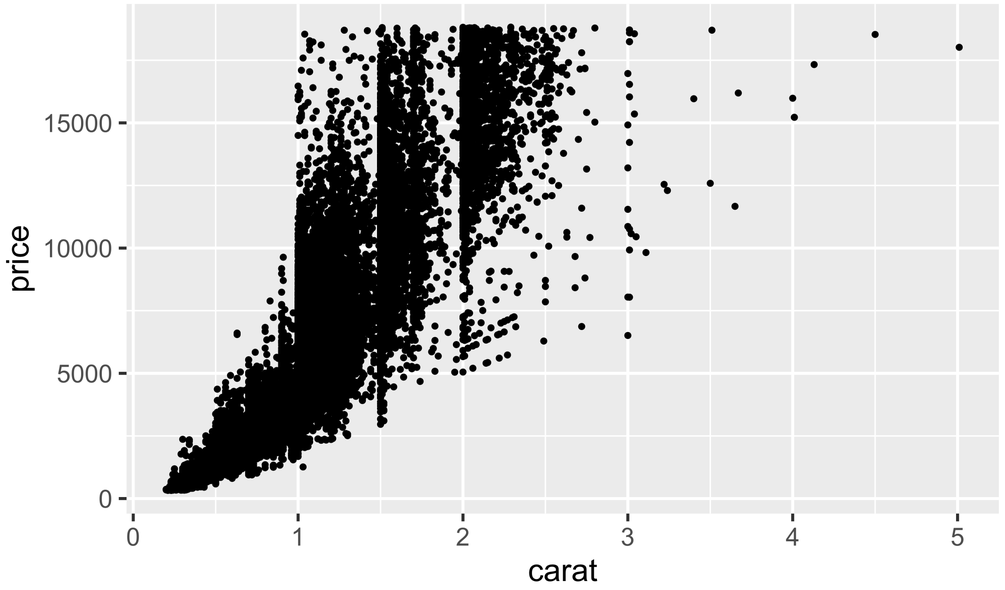
Geometrias para grandes bases de dados

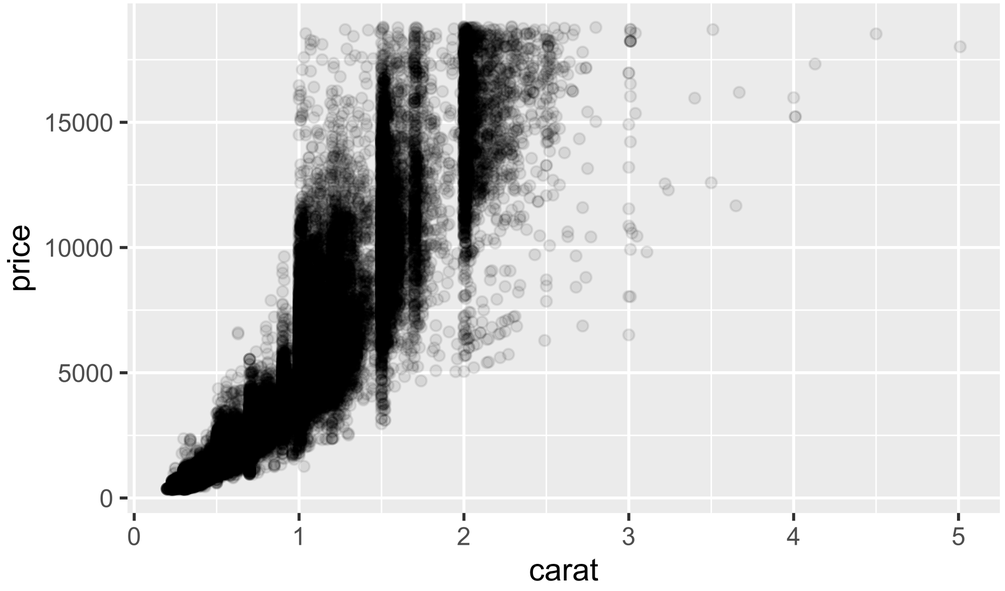
Big Data

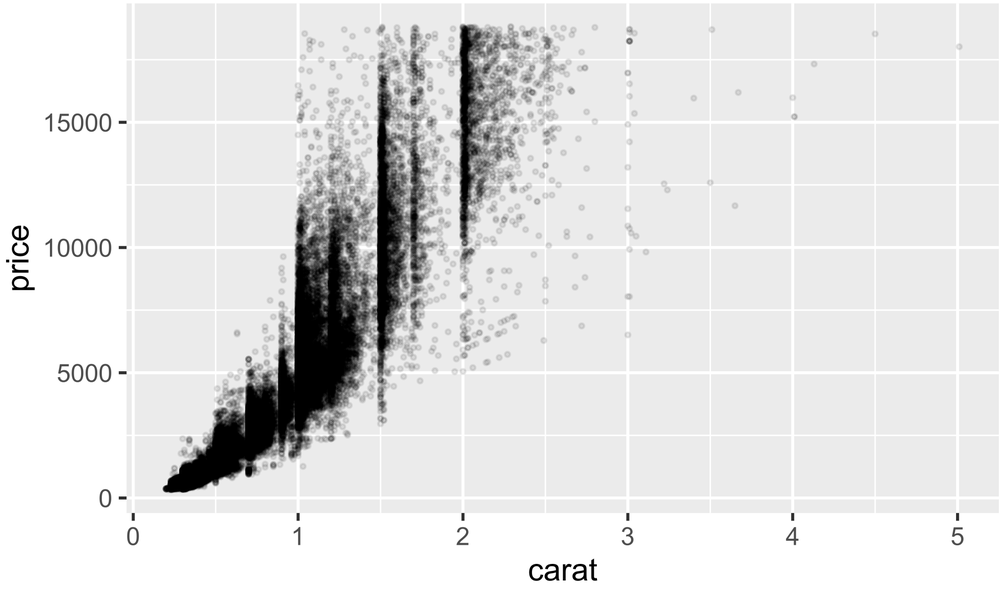


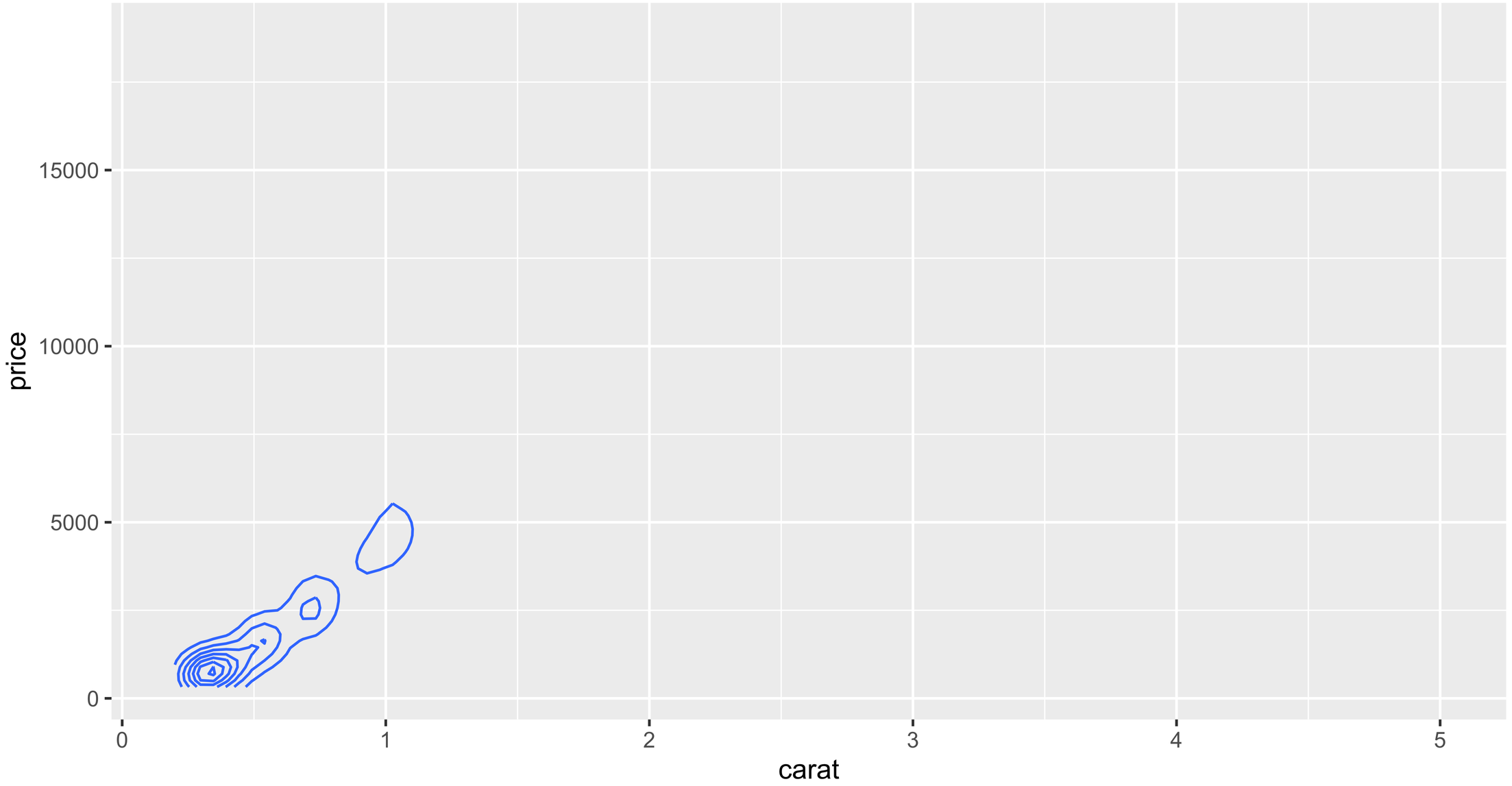

```
ggplot( diamonds, aes(x = carat, y = price) ) +
  geom_point( aes(color = cut) )
```


price
15000
10000
5000
0
0
1
2
3
4
5
carat

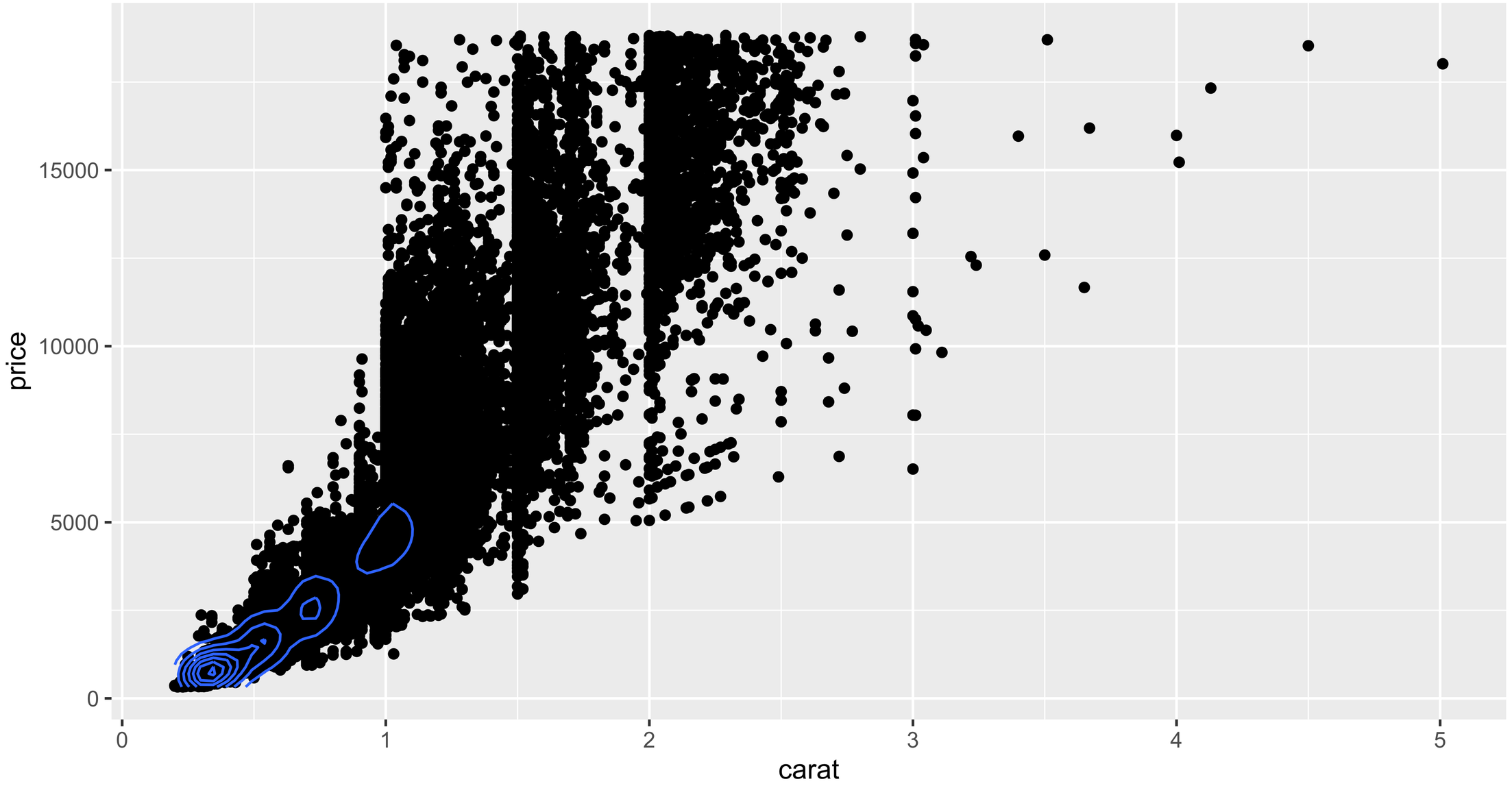

price
15000
10000
5000
0
0
1
2
3
4
5
carat


price
15000
10000
5000
0
0
1
2
3
4
5
carat

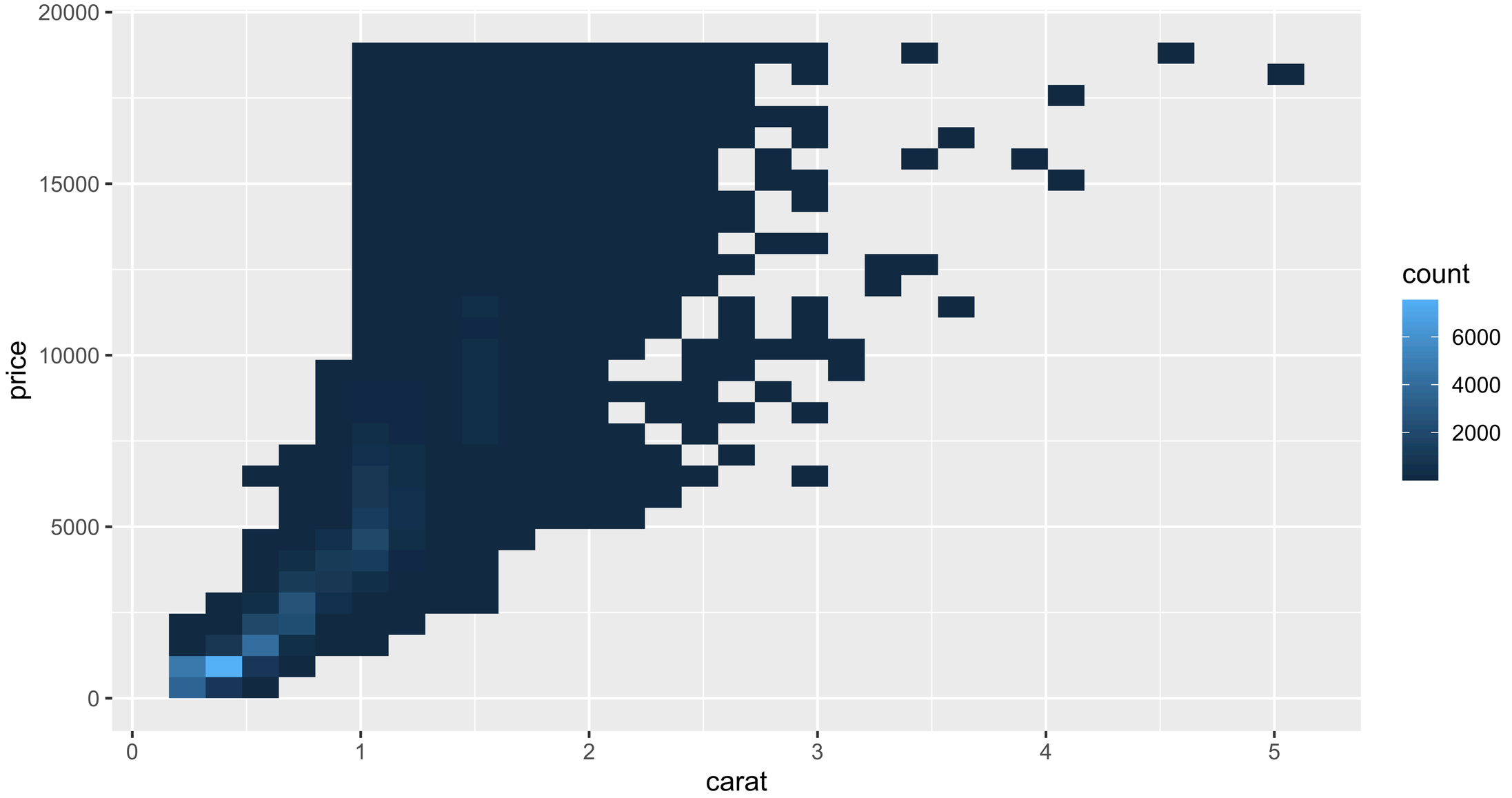

price
15000
10000
5000
0
0
1
2
3
4
5
carat



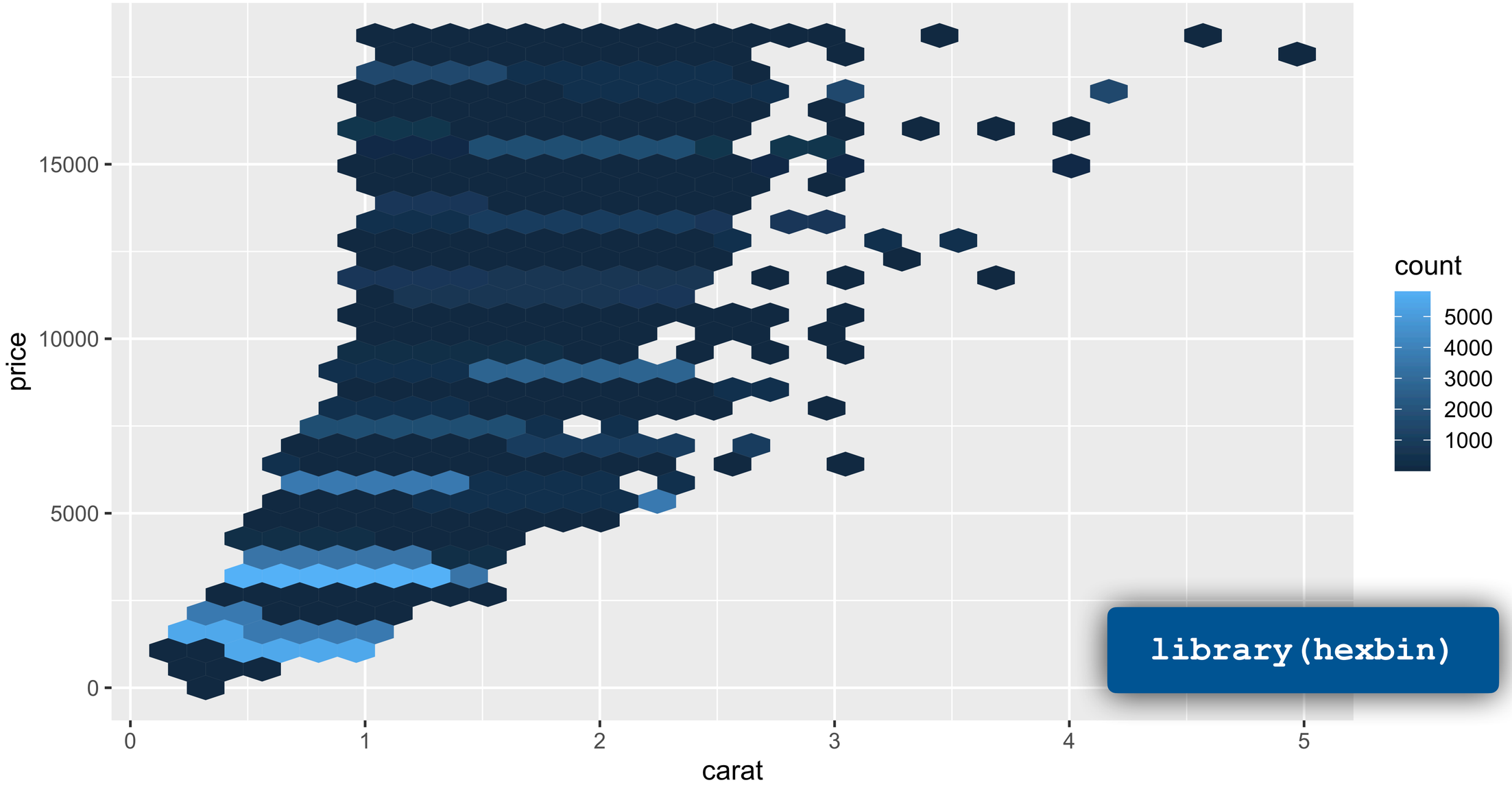

```
ggplot( diamonds, aes(carat, price) ) +
  geom_density2d()
```



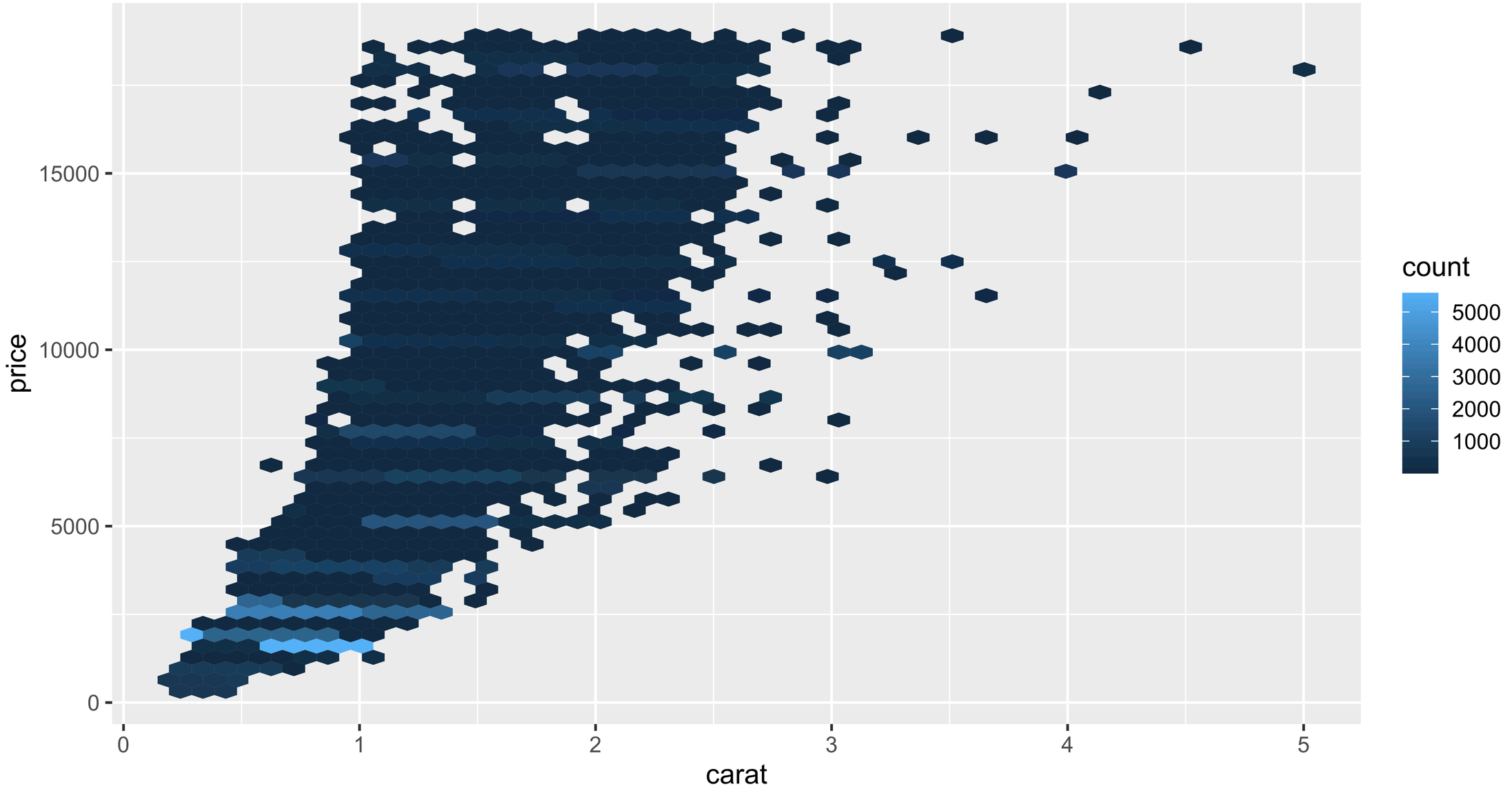

```
ggplot( diamonds, aes(carat, price) ) +
  geom_point() + geom_density2d()
```



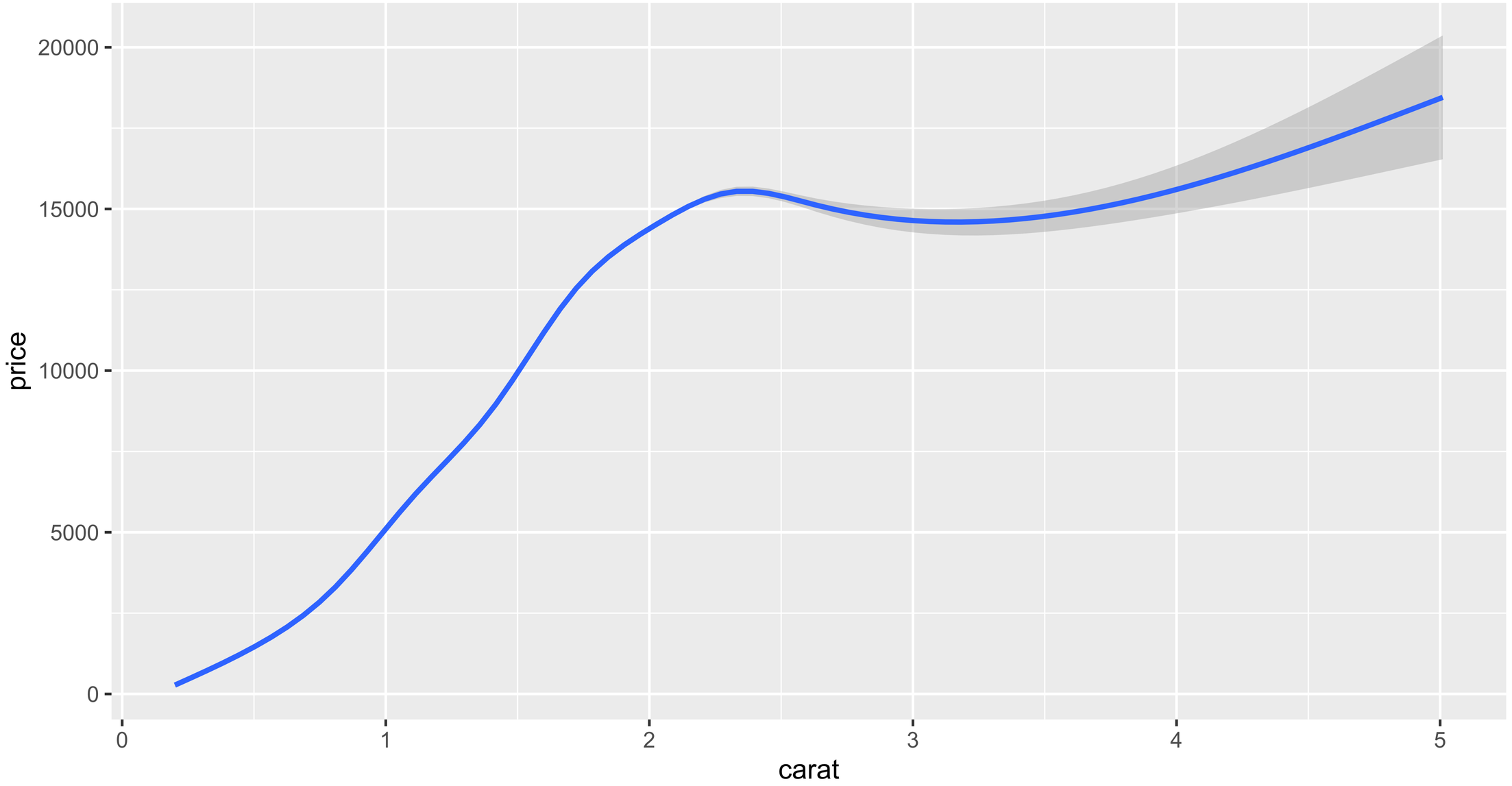

```
ggplot( diamonds, aes(carat, price) ) +
  geom_bin2d()
```



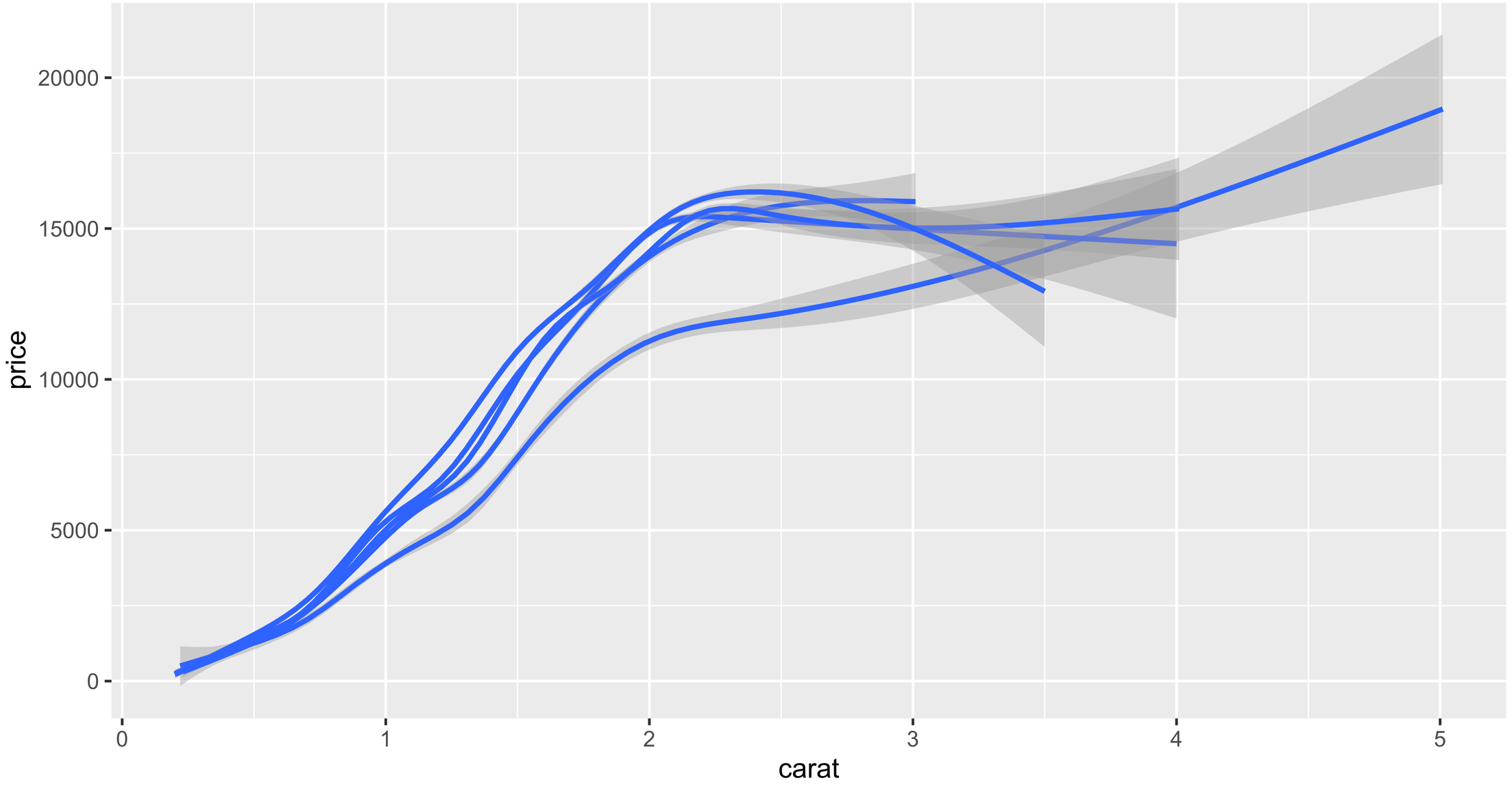

```
ggplot( diamonds, aes(carat, price) ) +
  geom_hex()
```



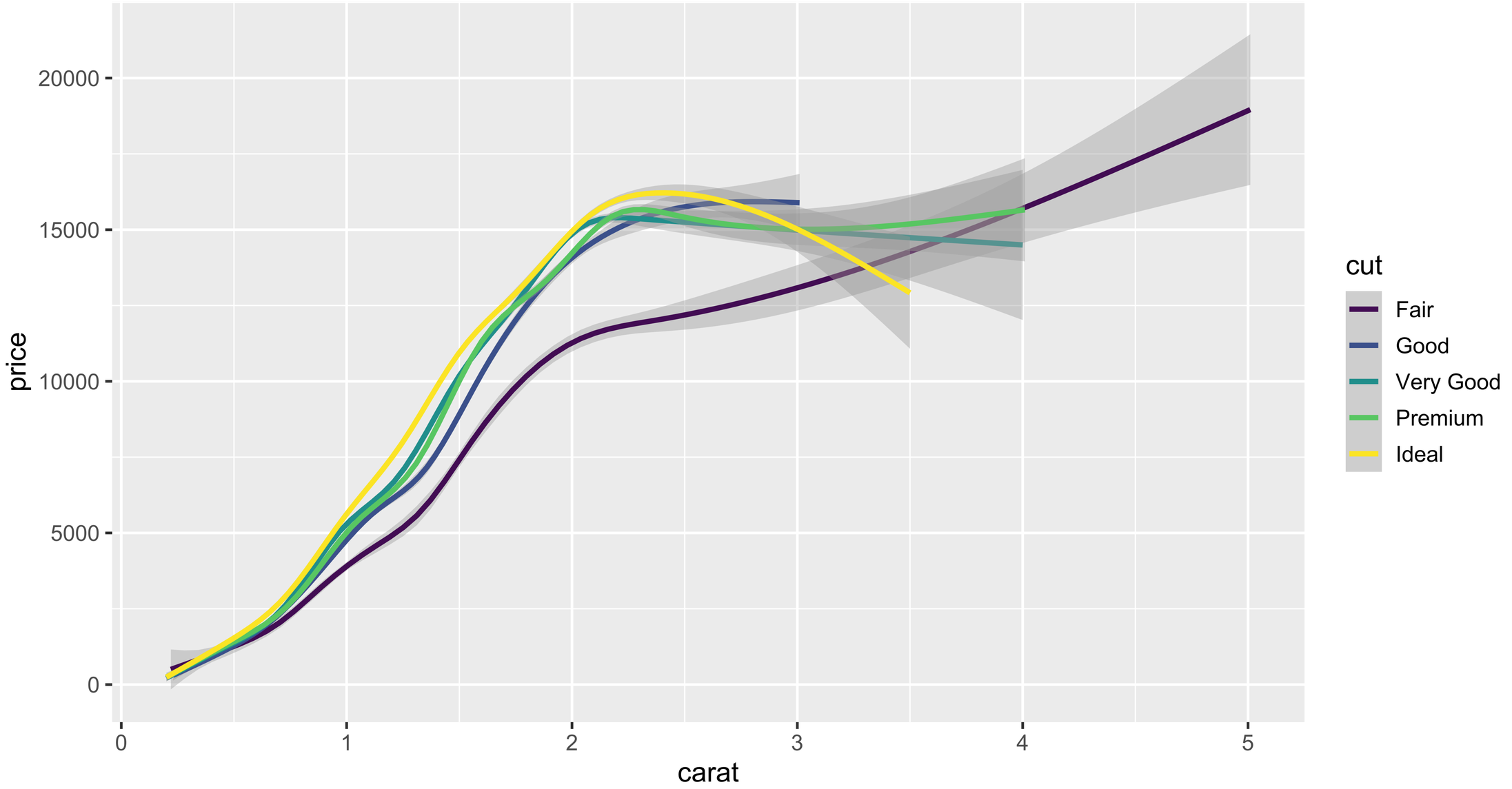

```
ggplot( diamonds, aes(carat, price) ) +
  geom_hex( bins = 50 )
```



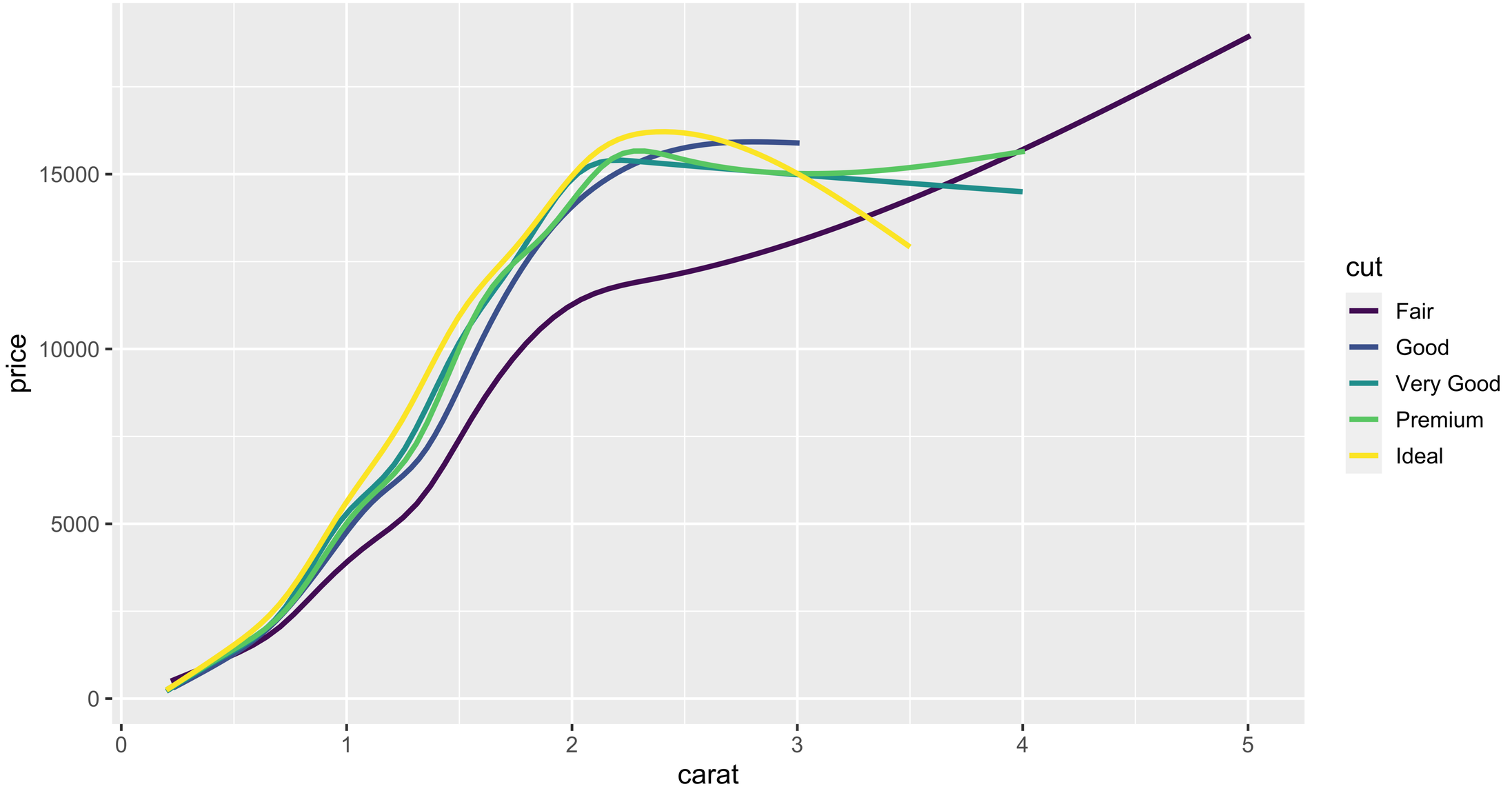

```
ggplot( diamonds, aes(carat, price) ) +
  geom_smooth()
```



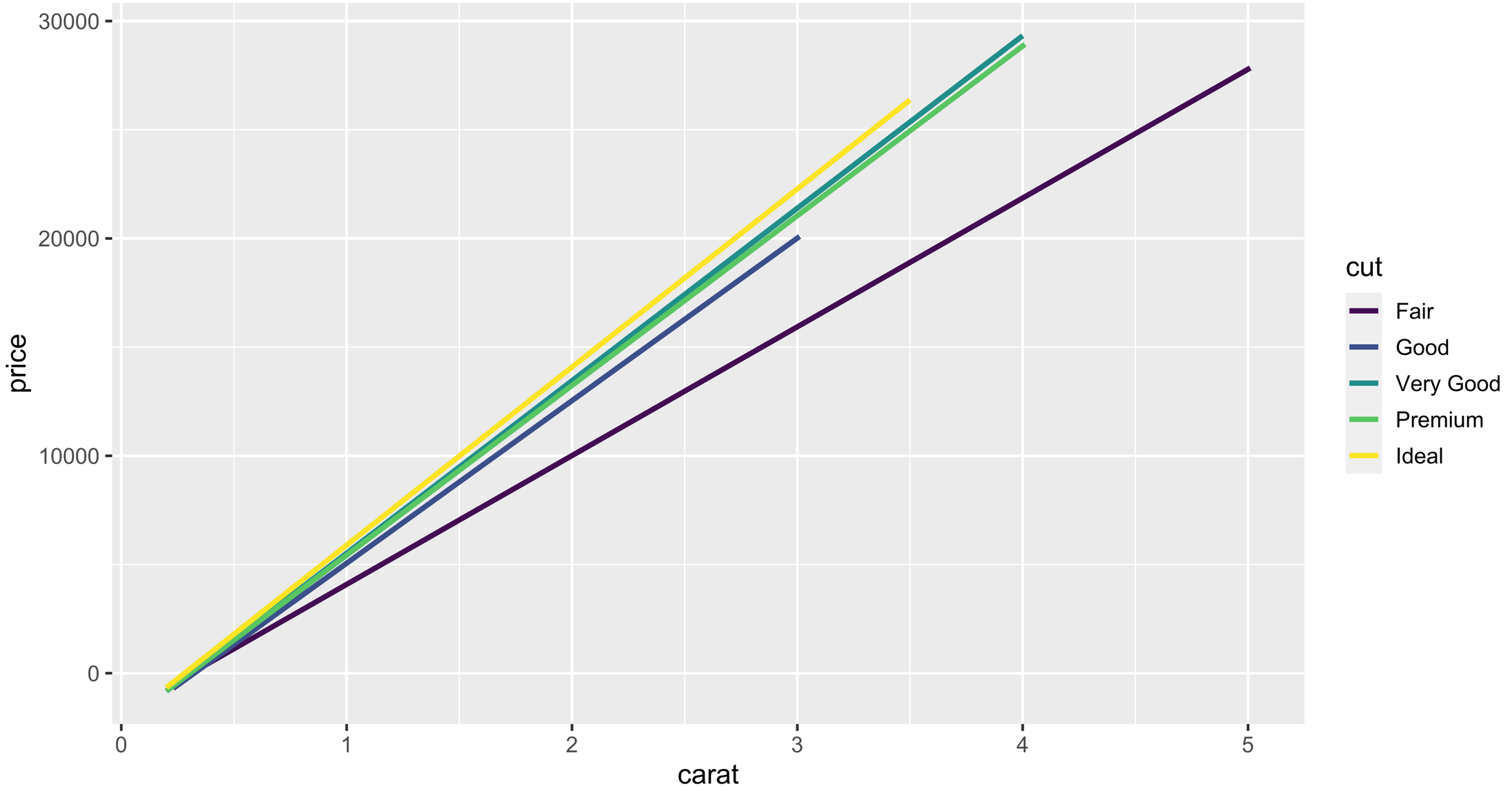

```
ggplot( diamonds, aes(carat, price) ) +
  geom_smooth( aes(group = cut) )
```




```
ggplot( diamonds, aes(carat, price) ) +
  geom_smooth( aes(color = cut) )
```




```
ggplot( diamonds, aes(carat, price) ) +
  geom_smooth( aes(color = cut), se = F )
```

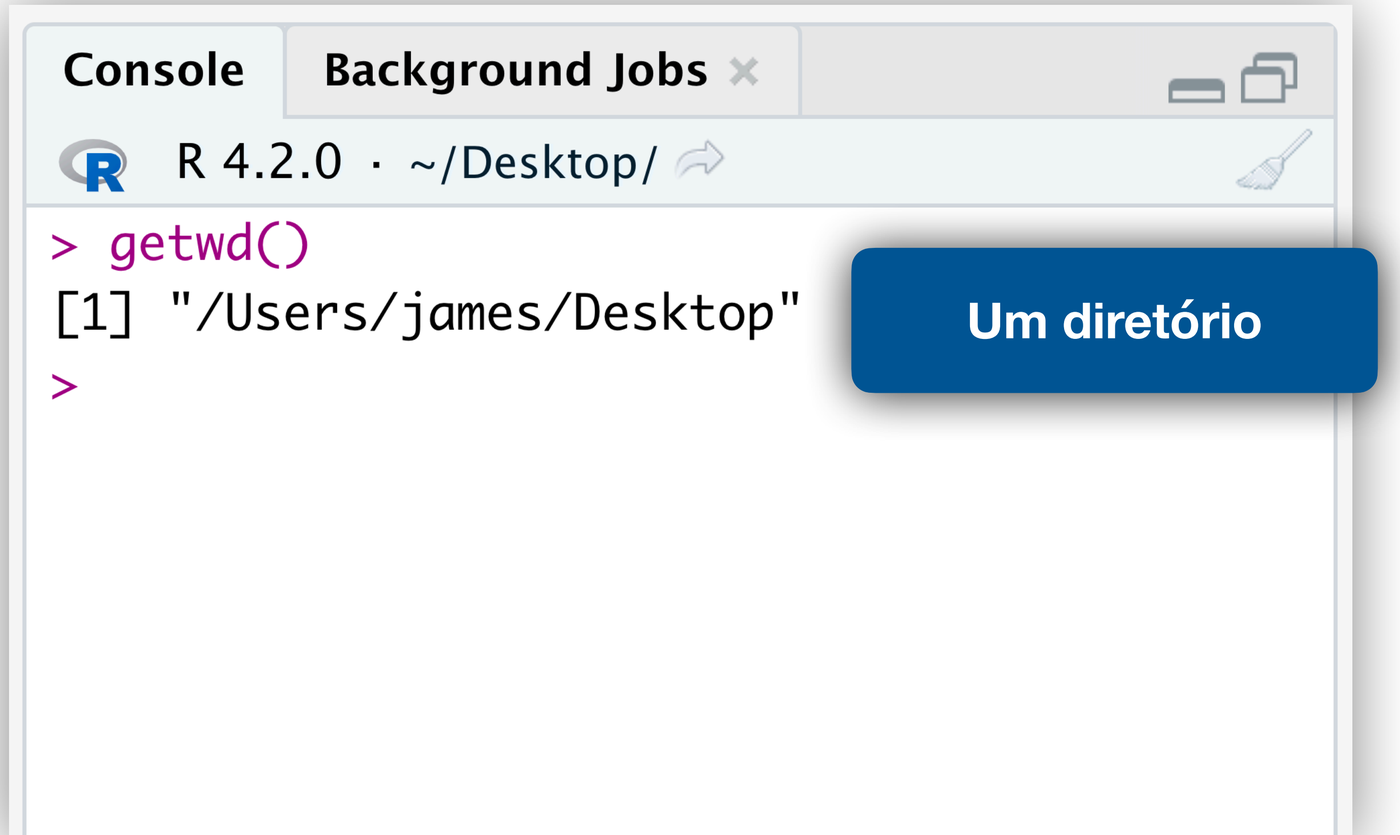



```
ggplot( diamonds, aes(carat, price) ) +
  geom_smooth( aes(color = cut), se = F, method = lm )
```

Salvando gráficos

SUA VEZ

- Execute o comando `getwd()`. Qual a saída?

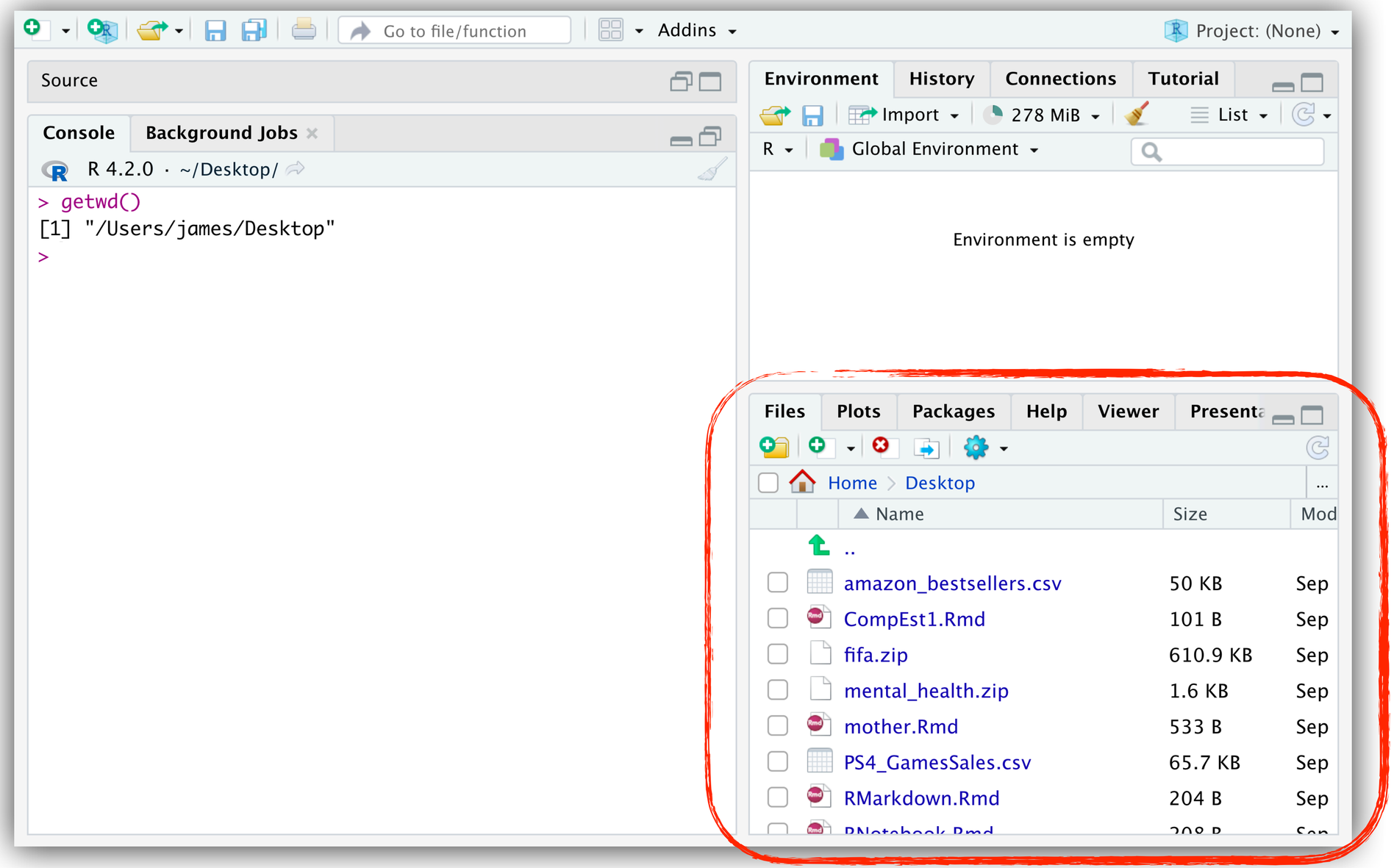


Working directory

Diretório de trabalho

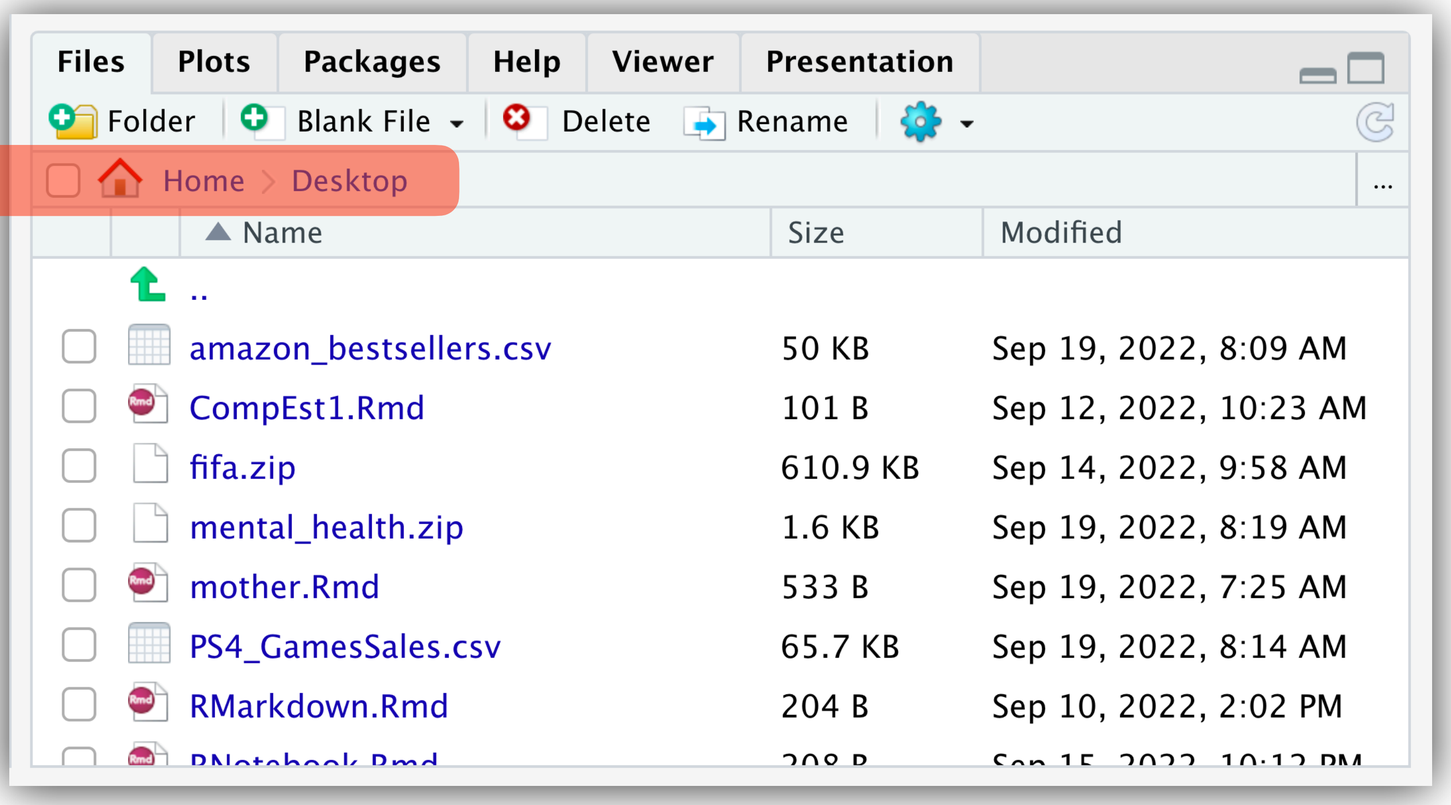
- Quando você iniciar o R, ele irá se associar a alguma pasta (diretório) do seu computador
- Esta pasta é conhecida como "working directory". Em português, "diretório de trabalho"
- Quando você salvar arquivos, o R irá salvá-los neste diretório
- Quando você quiser carregar arquivos, o R irá procura por eles, também, neste diretório (a não ser que você indique todo o caminho do arquivo)

A aba "Files" mostra todo o conteúdo do seu diretório de trabalho



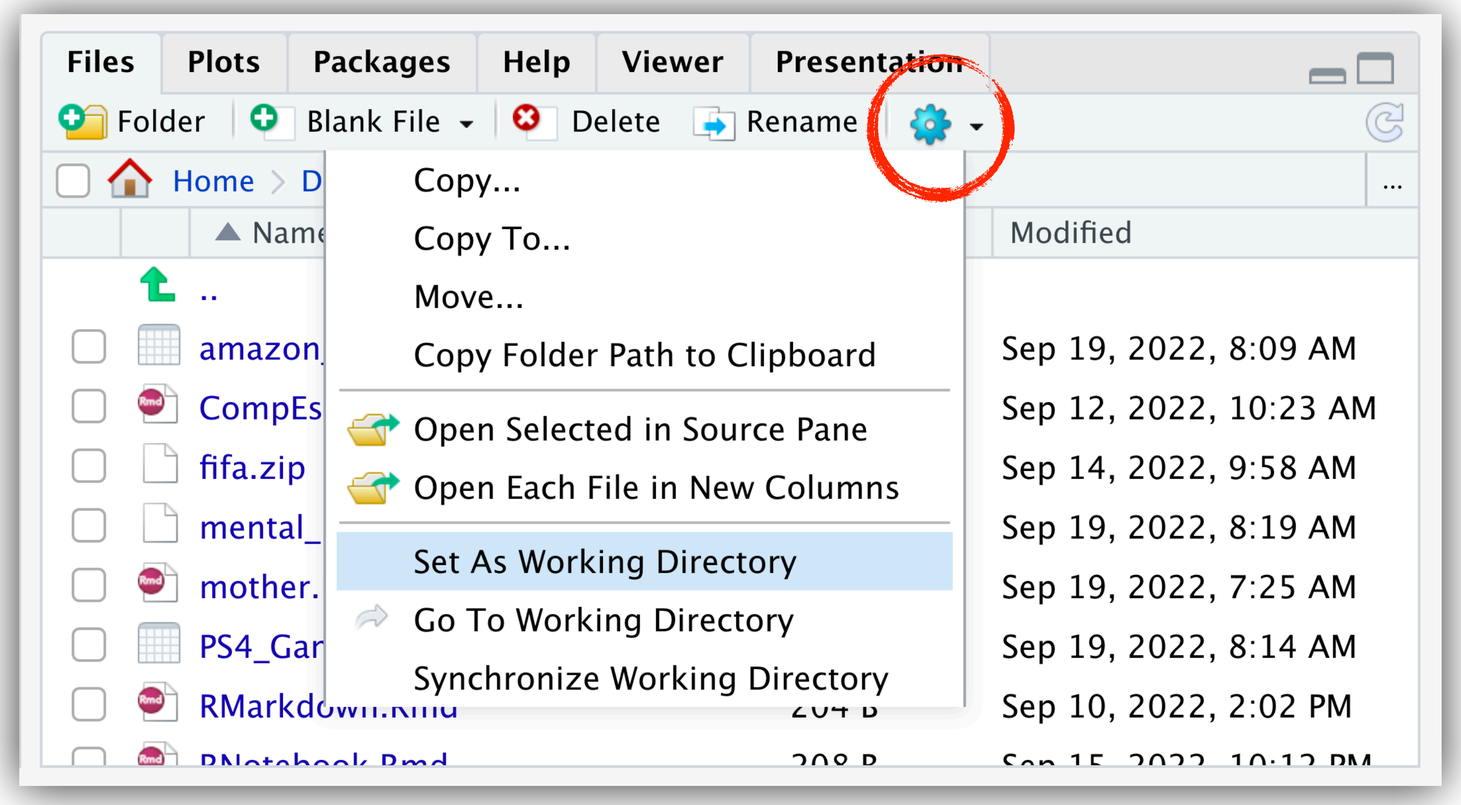
Alterando o diretório de trabalho

Opção 1: Navegue pelas pastas na aba *Files* e defina o diretório de trabalho



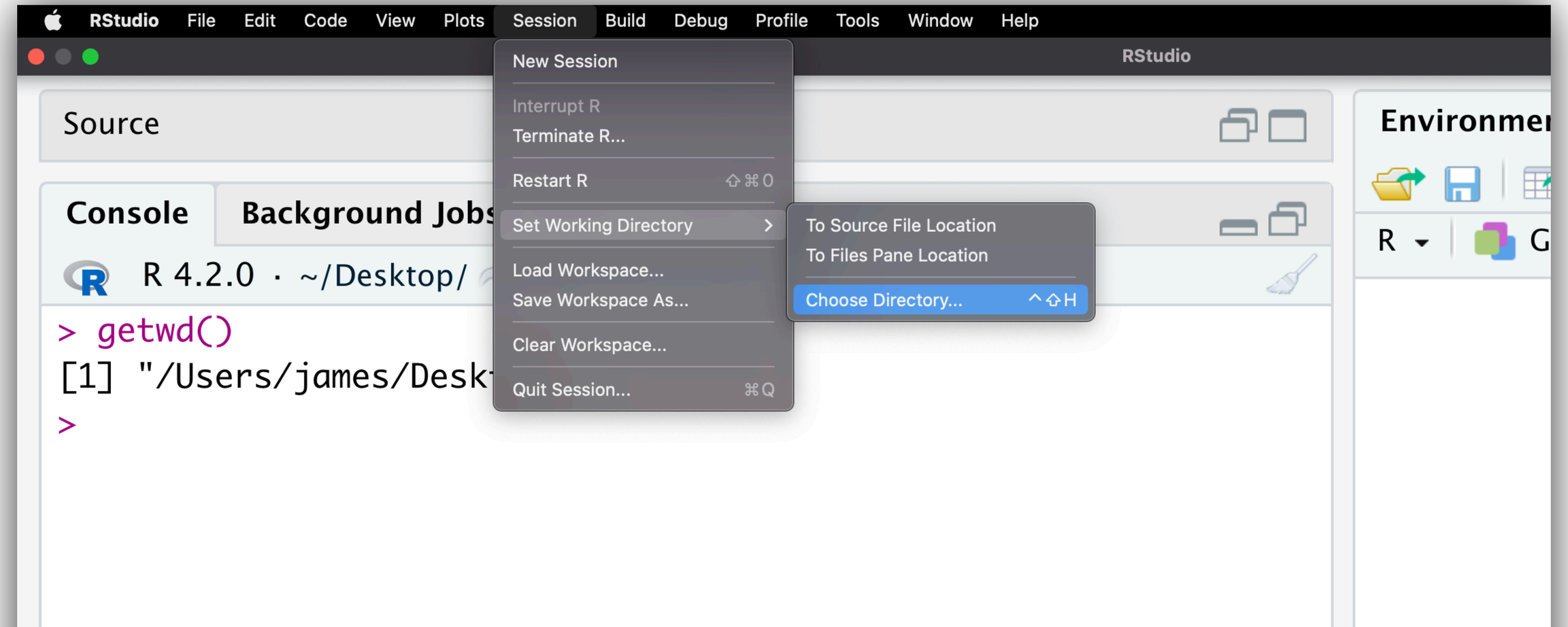
Alterando o diretório de trabalho

Opção 1: Navegue pelas pastas na aba *Files* e defina o diretório de trabalho



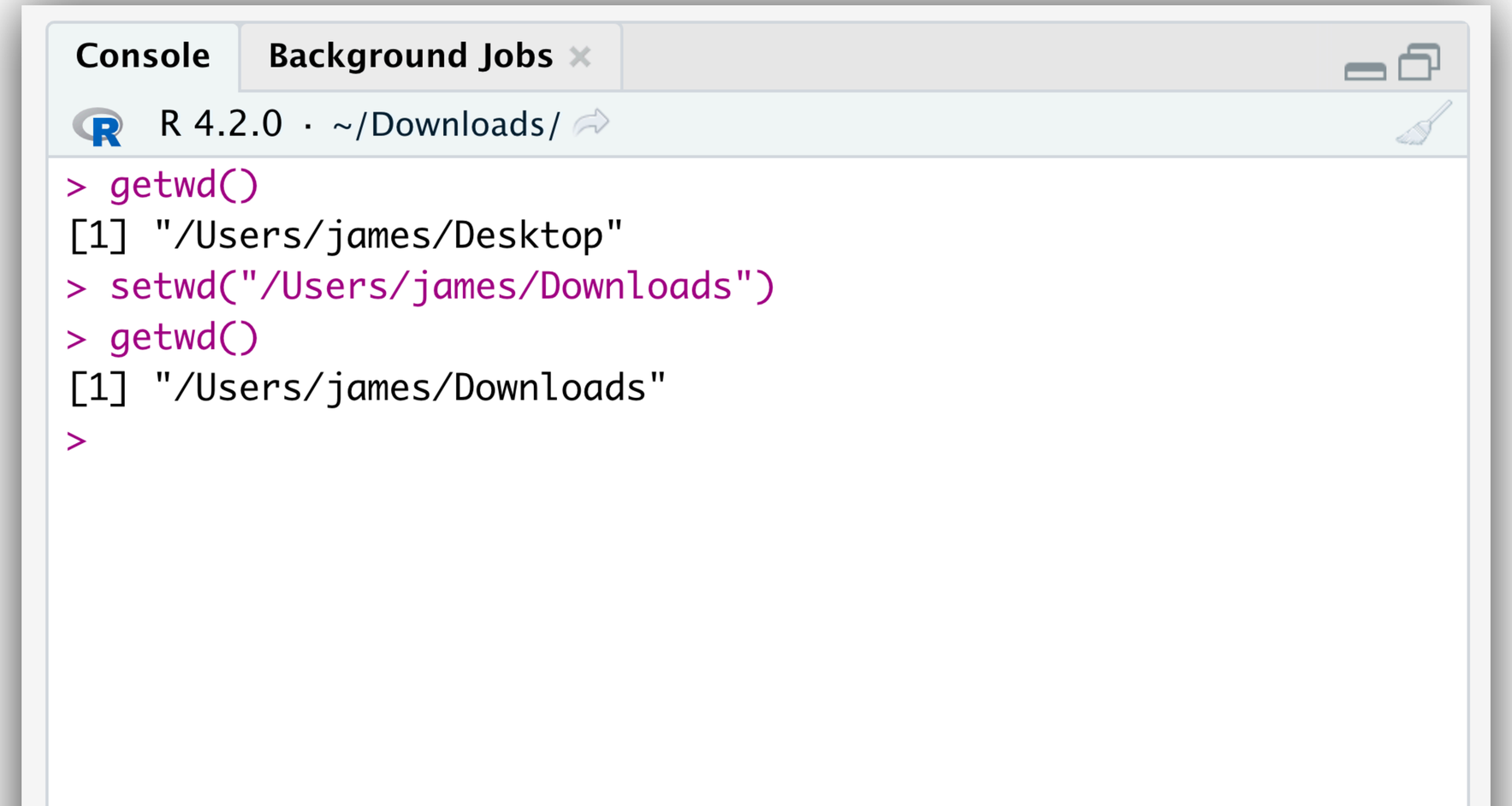
Alterando o diretório de trabalho

Opção 2: Altere na barra de tarefas



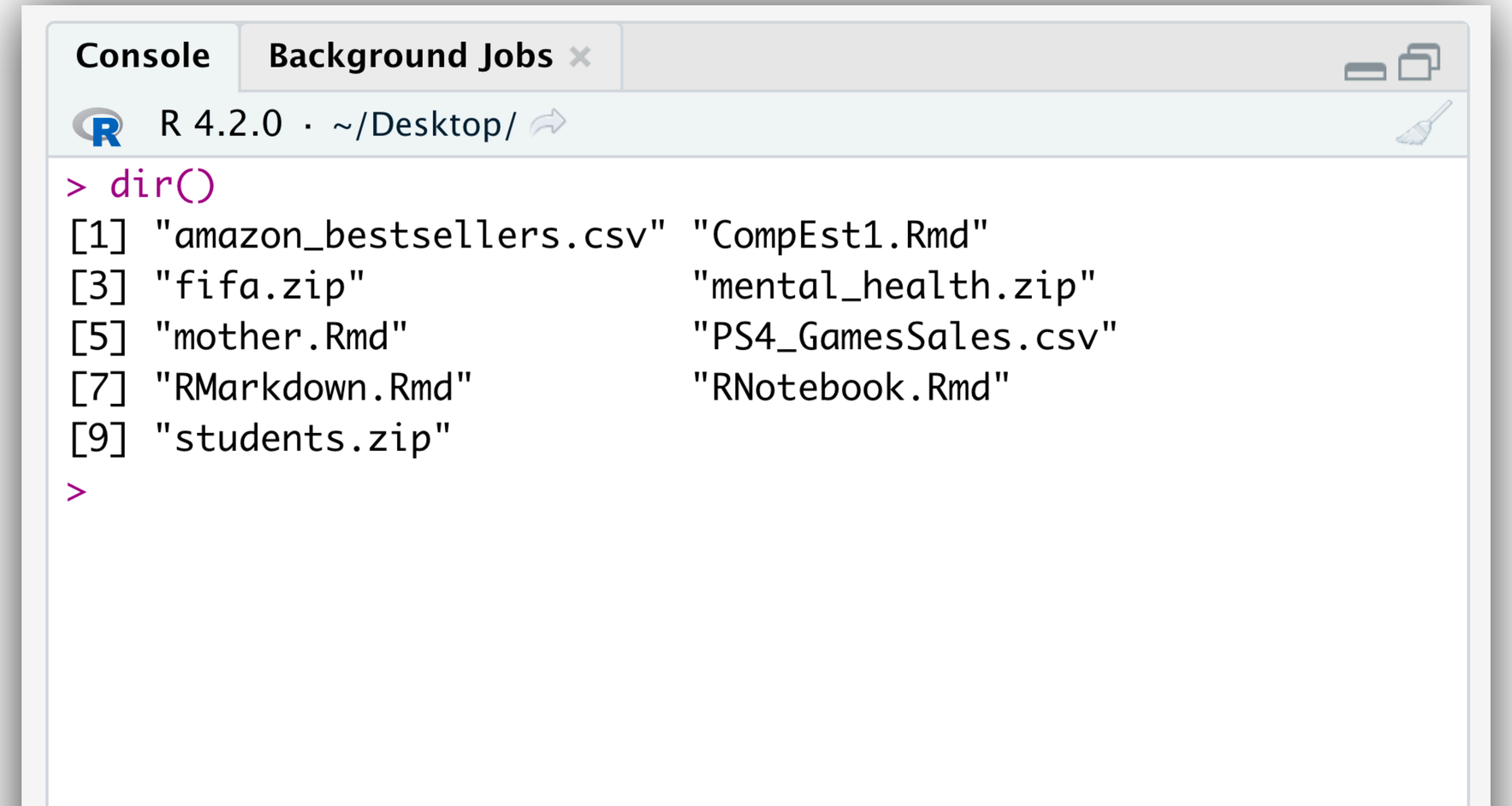
Alterando o diretório de trabalho

Opção 3: Utilize o comando `setwd()`



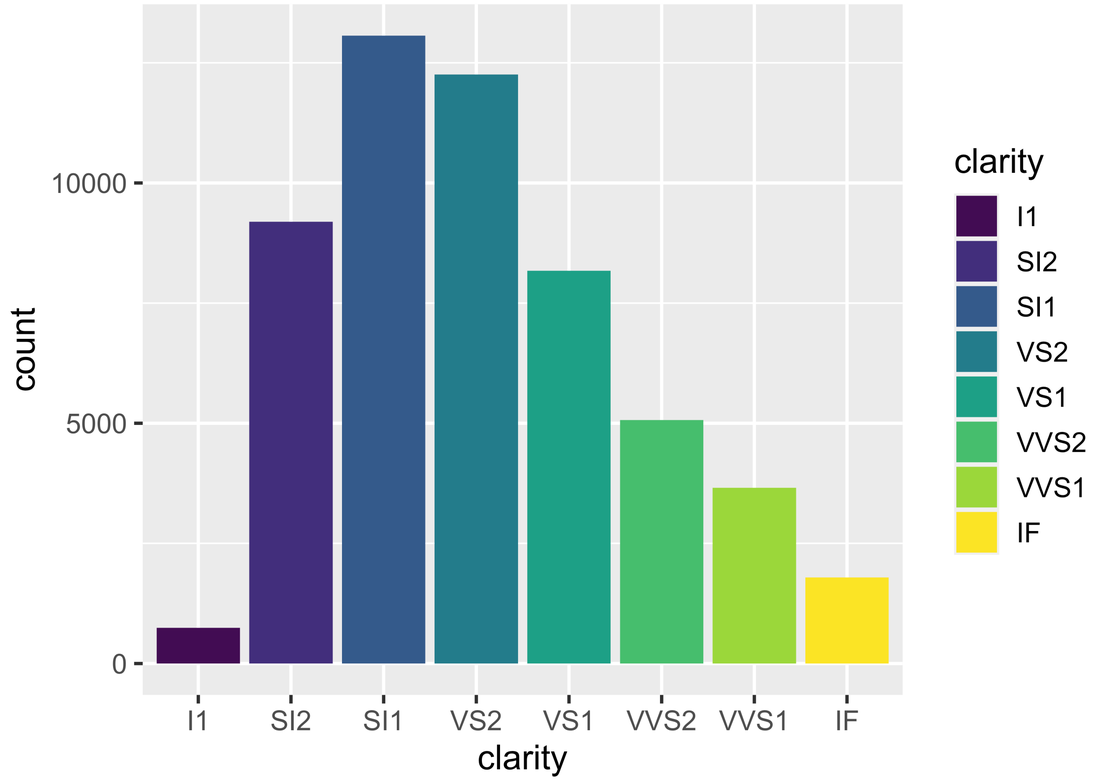
Arquivos

Utilize o comando `dir()` para ver os arquivos da pasta de trabalho



Salvando

```
my_plot <- ggplot( diamonds, aes(x = clarity) ) +
           geom_bar( aes(fill = clarity) )
```



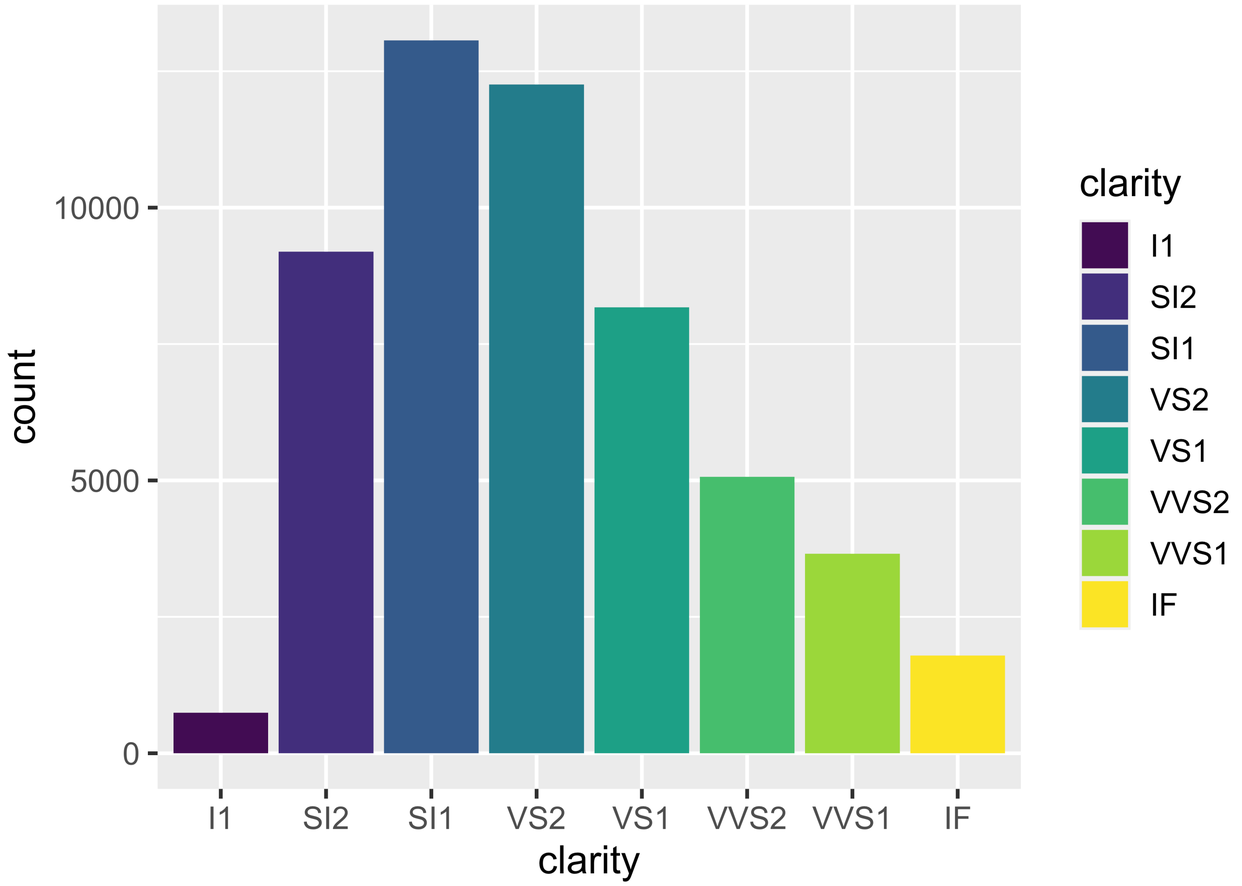
Salvando

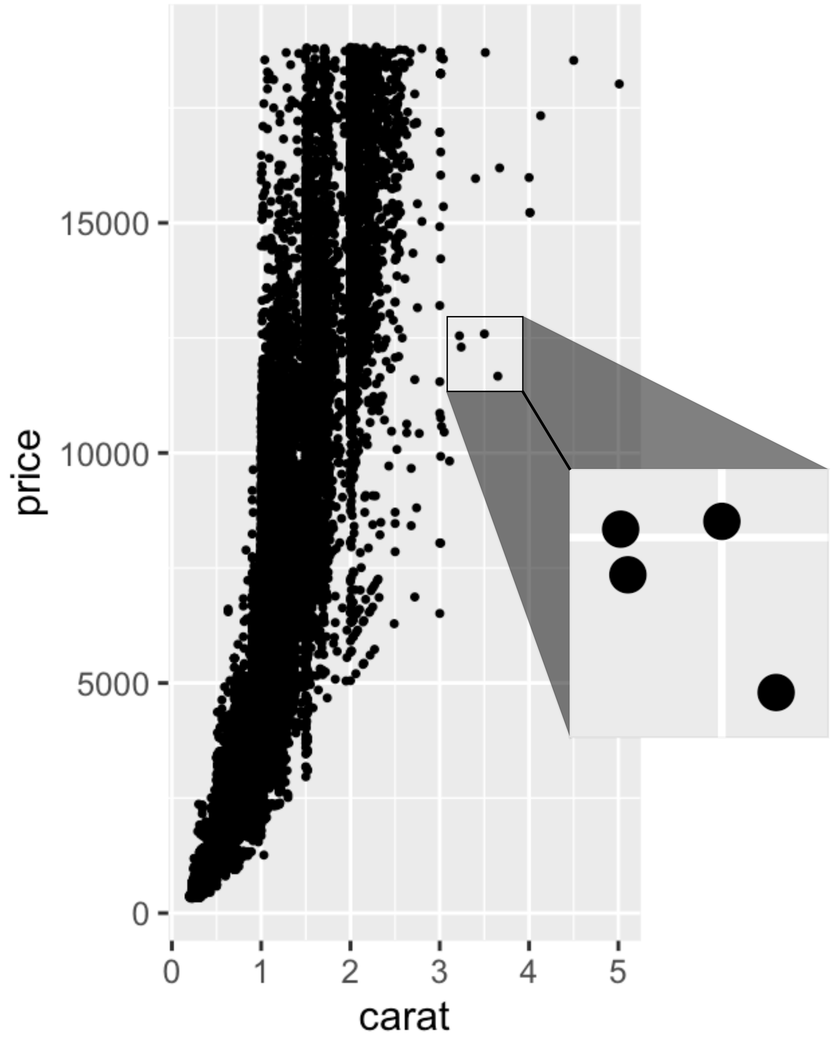
Utilize o tamanho da tela

```
ggsave("my_plot.pdf")
ggsave("my_plot.png")
```

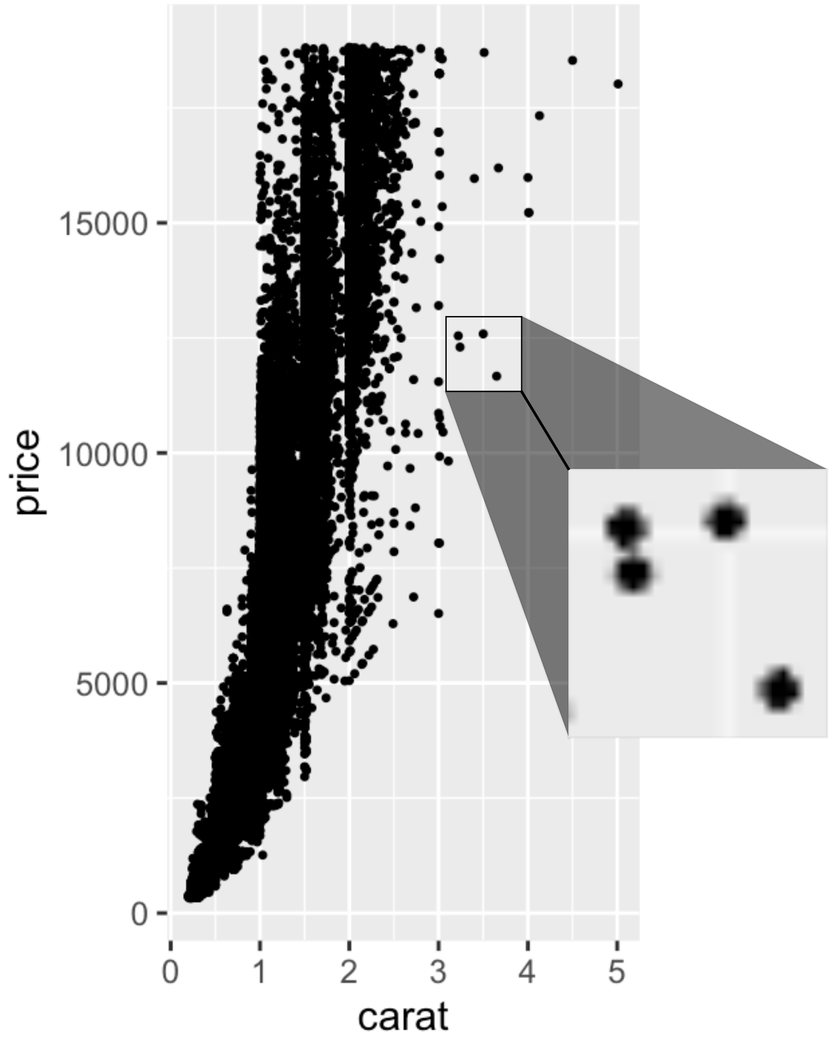
Escolha o tamanho

```
ggsave("my_plot.pdf",
       width = 6,
       height = 6)
```




price
15000
10000
5000
0
0
1
2
3
4
5
carat

PDF


price
15000
10000
5000
0
0
1
2
3
4
5
carat

PNG

PDF	PNG
Imagem vetorial (zoom infinito)	Imagem raster (pixels)
Ótima flexibilidade, porém pouca otimização sobre o tamanho do arquivo	Pouca flexibilidade, mas permite uma melhor otimização sobre o tamanho do arquivo